

Economic model predictive control for building multi-energy systems under evolving feed-in regulations: A comparative study with rule-based strategies

Niklas Kießling^{a,*}, Daniel Wolf^{b,a}, Fontina Petrakopoulou^a

^a Department of Energy Engineering and Climate Protection, Institute of Energy Engineering, Technische Universität Berlin, Marchstr. 18, 10587, Berlin, Germany

^b Solander GmbH, Rollbergstr. 28a, 12053, Berlin, Germany

ARTICLE INFO

Keywords:

Building energy management system
Economic model predictive control
Building multi-energy system
Load shifting
PV-battery system
Heat pump system

ABSTRACT

Advanced control strategies for building multi-energy systems are essential to manage renewable generation variability while maintaining grid stability and minimizing costs. This paper presents a comprehensive comparison between conventional rule-based control and economic model predictive control (EMPC) strategies for a residential building system comprising photovoltaic (PV) generation, battery storage, and an air-source heat pump with thermal storage, thereby combining detailed component modeling, multi-energy system optimization, and evaluation under recently introduced dynamic feed-in regulations. A mixed-integer linear programming framework is developed that incorporates part-load and temperature-dependent heat pump performance, battery charge acceptance, and equipment degradation costs. The control strategies are evaluated through time-resolved and annual analyses using real weather data and consistent demand profiles, considering two distinct regulatory scenarios for PV feed-in limitations: static power caps and dynamic zero-compensation periods during negative electricity prices. Results demonstrate that while an enhanced rule-based control strategy with forecast-based battery management and surplus-driven heat pump operation improves performance over the reference approach, EMPC achieves superior outcomes across all metrics. Compared to the reference rule-based control, EMPC reduces net annual costs by 37.5%, grid imports by 13.1%, battery throughput by up to 15.2%, heat pump cycling by 25.5%, and PV curtailment by 41.3%. EMPC's advantages stem from adaptively optimizing energy flows across multiple domains, strategically utilizing thermal energy storage to extend effective battery capacity, and minimizing conversion losses through intelligent power routing. These findings indicate that advanced predictive control becomes particularly advantageous under increasingly dynamic market and regulatory conditions.

1. Introduction

Buildings account for approximately 40% of EU-wide greenhouse gas emissions, making the decarbonization of the building sector essential for meeting the Paris Agreement targets [1–3]. Two significant trends are driving this transition: the increasing integration of distributed renewable energy sources, particularly photovoltaic systems, and the electrification of heating through heat pumps. Together, these developments create critical challenges for grid operation: managing the intermittency of renewable generation while accommodating additional electrical loads from heat pumps and the growing adoption of electric vehicles [4,5].

Two mutually reinforcing approaches have emerged to address these challenges: decentralized energy storage and load shifting. The integration of renewable energy sources calls for enhanced demand flexibil-

ity to match the volatile supply in electricity generation [6]. Decentralized multi-energy systems (MESSs) can offer various storage capabilities across multiple energy vectors (electrical, thermal, and chemical), creating synergistic opportunities through sector coupling [7]. By consuming or storing energy when and where it is available, these systems can significantly reduce conversion and transportation losses, thereby decreasing both operational costs and greenhouse gas emissions [8]. Further improvements may be achieved through the implementation of holistic and predictive energy management systems [9].

Control strategies for building MESSs can broadly be classified into two categories: rule-based and optimization-based. Rule-based control (RBC) represents the current state of practice in building energy systems, with distinct approaches established for different components. For battery energy storage systems, PV self-consumption maximization has emerged as the predominant strategy [10–13]. This approach follows a

* Corresponding author.

E-mail address: niklas.kiessling@tu-berlin.de (N. Kießling).

<https://doi.org/10.1016/j.enbuild.2026.118017>

Received 25 March 2026; Received in revised form 12 July 2026; Accepted 25 July 2026

Available online 29 July 2026

0378-7788/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Symbols, indices and acronyms*Latin symbols*

A	Area
b	Model coefficient
C	Heat capacity
c	Cost coefficient
d	Duration
E	Energy
J	Cost function
M	Big-M constant
P	Electric power
$\dot{Q}$	Heat flow rate
R	Request
T	Time at end of prediction horizon
t	Time
U	Overall heat transfer coefficient
u	On/off variable
v	Start-up variable

Greek symbols

Δ	Difference
ϵ	Slack variable
η	Efficiency
ϑ	Temperature

Indices

AC	Alternating current
Bat	Battery
CA	Charge acceptance
CC	Charge controller
ch	Charging
Cur	Curtailement
DC	Direct current
dch	Discharging
ex	Export
Grid	Electric grid
HE	Heating element
HP	Heat pump
im	Import
Inv	Inverter
Load	Household electrical load
max	Maximum
min	Minimum
nom	Nominal
Stby	Standby consumption

Acronyms

COP	coefficient of performance
DHW	domestic hot water
EMPC	economic model predictive control
ERBC	enhanced rule-based control
MES	multi-energy system
MILP	mixed-integer linear programming
MPC	model predictive control
PI	proportional-integral
PV	photovoltaic
RBC	rule-based control
RRBC	reference rule-based control
SH	space heating
SoC	state of charge
TES	thermal energy storage

before being fed into the grid. Similarly, when electricity consumption exceeds PV generation, the battery is first fully discharged before grid power is utilized. This simple yet effective operating strategy has guaranteed economically secure storage utilization within Germany's regulatory framework over recent years, particularly under the feed-in power limitation at 70% of installed peak capacity that was in effect from 2012 to 2023 [14].

However, regulatory changes in Germany taking effect in February 2025, including the general reduction of feed-in power limitations to 60% or, alternatively, the elimination of feed-in remuneration during periods of negative day-ahead spot market prices for new installations without and with smart meters, respectively, are shifting the economic landscape [15]. This development reflects a broader European move toward stronger market-based integration of renewable electricity [16, 17] and is therefore likely to increase the relevance of forecast-based charging strategies that avoid curtailment while preserving maximum self-sufficiency [10,18–20].

For space heating (SH) and domestic hot water (DHW) provision with variable compressor speed heat pumps, current control strategies predominantly rely on two separate control loops [21–24]. DHW preparation typically employs a two-point controller with hysteresis and fixed compressor speed to maintain weekday and time-of-day dependent temperature setpoints, with this function prioritized over SH. SH control generally implements an outdoor temperature-dependent supply temperature setpoint (heating curve) maintained by a proportional-integral (PI) controller that varies compressor speed. Additionally, an upper hysteresis limit is often utilized to extend runtime at minimum compressor speed. These control approaches typically incorporate restrictions on minimum runtime and maximum start-ups per hour to extend compressor service life. A back-up electric heater with discrete power levels is commonly activated incrementally, e.g., if the setpoints are under-shot by a defined threshold and/or remain unmet after specified time intervals.

This control strategy primarily aims to minimize energy consumption which, assuming constant energy prices, also ensures low operational costs. However, with the widening gap between grid purchase costs and feed-in revenues, both reactive and predictive strategies for thermal storage of excess PV electricity—such as temporarily increasing temperature setpoints or adjusting power levels—are becoming increasingly attractive [19,22,25,26].

In large-scale power systems, unit commitment and dispatch optimization aims to determine which generators to operate when and at what levels to reliably meet electricity demand at the lowest total cost, subject to operational and regulatory limitations. The most widely adopted formulation for these problems is mixed-integer linear programming (MILP), which explicitly captures both the binary on/off decisions and continuous dispatch variables while handling complex constraints and leveraging the robustness and efficiency of modern solvers [27,28].

Mathematical optimization techniques developed in this domain promise substantial advantages for building MESs operation [29–31]. The numerous possible energy flows resulting from the growing number of interconnected components, combined with highly user-specific and/or site-specific demand and generation patterns, create a level of complexity that exceeds the capabilities of RBC strategies. This transition is further enabled by the increasing accuracy of statistical forecast models, particularly those using machine learning approaches [32–34].

A critical consideration in control applications is that optimization results need to be continuously adapted to changing conditions, for example, to respond effectively to model and prediction errors, requiring strategies that can iteratively refine solutions as new information becomes available. Rolling horizon and receding horizon optimization approaches address the dynamic nature of energy system control by repeatedly solving optimization problems over a moving time window. These two concepts are often used interchangeably in literature and share fundamental principles: they solve problems over a limited future

straightforward priority scheme: excess PV generation beyond immediate electricity consumption is first directed to fully charge the battery

horizon, implement part of the resulting plan or policy immediately, and then shift the horizon forward to re-optimize.

Despite these similarities, subtle distinctions exist between the approaches. Terminology often varies by research field, with operations research typically favoring “rolling horizon” while control theory uses “receding horizon control” (also known as model predictive control (MPC)). The frequency of updates also differs, with receding horizon approaches typically updating at nearly every time step for real-time closed-loop control, while rolling horizon methods may incorporate coarser intervals encompassing multiple time steps. Beyond these structural differences, control theory distinguishes MPC approaches based on their objective functions. Standard MPC primarily aims to stabilize or track desired system trajectories while respecting operational constraints, typically employing quadratic cost functions that provide smooth penalties for deviations from reference setpoints. EMPC, by contrast, directly optimizes cost or profit objectives to enhance the system’s overall economic performance utilizing the flexibility of controlled variables within their admissible operating ranges and tends to favor linear cost functions that more accurately represent real-world economic relationships such as energy prices and operational costs.

Table 1 systematically compares related work along the modeling, optimization, and regulatory aspects addressed in this study. The examination reveals that existing studies either sacrifice model fidelity for system breadth or vice versa, while systematic comparisons against rule-based baselines accounting for contemporary regulatory conditions also remain scarce. To fill this gap, the present study offers three primary contributions:

- **Detailed cross-domain EMPC framework:** A MILP-based EMPC formulation that simultaneously captures electrical and thermal subsystems, including temperature-dependent heat pump performance, battery charge acceptance, and degradation-aware cost modeling.
- **Realistic benchmarking against advanced heuristics:** A structured and fair comparison against both conventional and forecast-enhanced RBC, reflecting practical implementations rather than simplified baselines.
- **Evaluation under emerging regulatory conditions:** A systematic analysis of control performance under both static feed-in caps and dynamic zero-compensation periods during negative electricity prices.

Table 1
Related work on (E)MPC of building MESs including heat pumps.

Reference	Generation/Load				Heat pump						Battery	Feed-in		(E)MPC setup				RBC	Simulation				
	PV	Household	SH	DHW	SH and DHW capable	Variable-speed compressor	Part-load dependent COP	Supply-temperature dependent COP	Temperature dependent power limits	Back-up heating element	Cycling cost	Charge acceptance	Throughput cost	Static feed-in limit	Zero-remuneration during neg. prices	High resolution ($\Delta t \leq 15$ min)	Prediction horizon ≥ 24 h	Single-stage (holistic)	No post-processing needed	RRBC	ERBC	Full year(s)	Coherent input data
Baumann et al. [35]	✓	✓	✓			✓						-	-			?	✓	✓	✓	✓	✓	✓	✓
Betzold et al. [36]	✓	✓	✓	✓	(✓) ^a	✓	✓						✓			✓	✓	✓	✓	✓	✓	✓	?
Bianchini et al. [37]	✓	✓	✓			✓										✓	✓	✓	✓	✓	✓	✓	
Corinaldesi et al. [38]	✓	✓	✓	✓	(✓) ^b	✓				(✓) ^b						✓	✓	✓	✓	✓	✓	✓	
Fischer et al. [21]	✓	✓	✓	✓	✓	✓	✓	(✓) ^c	(✓) ^d	✓	(✓) ^e	-	-			?	✓	✓	✓	✓	✓	✓	
Fitzpatrick et al. [39]			✓			✓											✓	✓	✓	✓	✓	✓	
Gelleschus and Bocklisch [40]	✓	✓		✓		✓								✓	-	✓	✓	✓	✓	✓	✓	✓	?
Harb [41]	✓	✓	✓	✓	✓			(✓) ^c		✓	(✓) ^e	✓				✓	✓	✓	✓	✓	✓	✓	
Kuboth et al. [42]	✓	✓	✓	✓	✓	✓		✓	✓	✓		✓			✓	✓	✓	✓	✓	✓	✓	✓	
Langer and Volling [43]	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	?	
Lee et al. [5]			✓			✓	✓					-	-	-	-		✓	✓	✓	✓	✓	✓	
Liu et al. [44]	✓	✓	✓			✓	✓	(✓) ^c	✓				✓			✓	✓	✓	✓	✓	✓	✓	
Löhr and Mönnigmann [45]	✓	✓	✓			✓				✓	(✓) ^e					✓	✓	✓	✓	✓	✓	✓	
Maier et al. [46]			✓			✓		✓				-	-	-	-		✓	✓	✓	✓	✓	(✓) ^f	(✓) ^g
Murray et al. [47]	✓	✓	✓	✓	✓	✓				✓						✓	✓	✓	✓	✓	✓	✓	✓
Rafii-Tabrizi et al. [48]	✓	✓	✓	✓	✓			✓	✓							✓	✓	✓	✓	✓	✓	✓	(✓) ^h
Sawant et al. [49]			✓							✓	(✓) ^e	-	-	-	-	✓	✓	✓	✓	✓	✓	✓	(✓) ^g
Schütz et al. [50]			✓					✓	✓			-	-	-	-	✓	✓	✓	✓	✓	✓	✓	(✓) ^g
Tang et al. [51]				✓								-	-	-	-	✓	✓	✓	✓	✓	✓	?	?
Tomás et al. [52]			✓					(✓) ^c				-	-	-	-	✓	✓	✓	✓	✓	✓	(✓) ^h	(✓) ^h
Verhelst et al. [53]			✓			✓	✓	✓	✓			-	-	-	-	✓	✓	✓	✓	✓	✓	✓	✓
This study	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

- = Battery and/or PV feed-in not part of study.

? = Information not reported or unclear from the publication.

(✓) = Partial or conditional fulfillment, see footnotes.

^a Different heat pumps are used for SH and DHW.

^b Heat pump serves SH only; DHW is provided by the heating element.

^c COP dependency modeled in simulation only; fixed a priori estimate used in the optimization.

^d Heat pump minimum power constraint enforced via post-processing correction of optimization results.

^e Cycling implicitly limited through minimum runtime constraints (no explicit cost term).

^f Annual results extrapolated from a set of typical days; no continuous full-year simulation.

^g Generation and load profiles based on a synthetic test reference year, combined with historical day-ahead prices.

^h Control strategies evaluated experimentally over non-identical time periods.

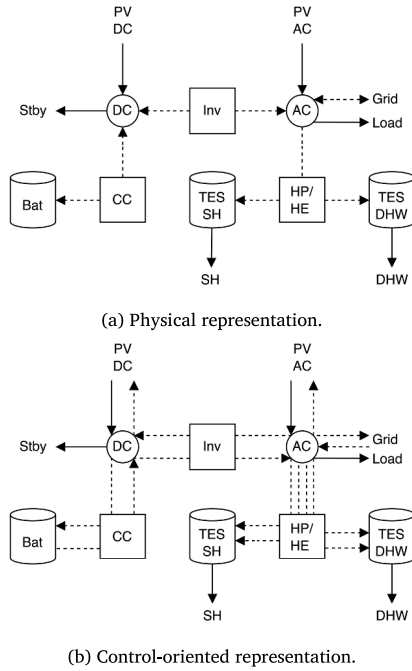


Fig. 1. Building MES structure.

2. Methodology

2.1. Overview

Fig. 1a illustrates the physical archetypal building MES layout considered in this study, comprising a grid-parallel DC- and/or AC-coupled PV-battery system, including charge controller (CC) and inverter (Inv). SH and DHW demands are met by a variable-speed air-source heat pump (HP) that is equipped with a three-stage electric back-up heating element (HE) and buffered by two thermal energy storage (TES) using water tanks. Solid arrows in the figure denote energy flows that are exogenous to the supervisory controller, while dashed arrows highlight the manipulated variables that can be accessed in real installations, e.g., via standard Modbus-TCP interfaces provided by the respective equipment manufacturers [54–56].

While PV generation remains inherently non-dispatchable due to solar irradiance variability, electrical standby (Stby) and household (Load) consumption, as well as thermal demands SH and DHW, are equally considered non-shiftable in this analysis. This assumption reflects the practical limitations of integrating sophisticated control systems for heat distribution and household appliances, as such integration would require additional measurement technology and/or substantial user interaction [57,58]. Moreover, this approach prevents potential comfort losses and acceptance problems.

2.2. Model

In order to enable simulation and optimization of control variables, the physical system representation is transformed into a linear mathematical model (as illustrated in Fig. 1b) that captures the essential behavior while maintaining computational tractability. Therefore, single bidirectional energy flows are decomposed into separate unidirectional flows as required by the optimization formulation. Specifically, the inverter and charge controller flows are split by direction to assign dedicated efficiencies, grid exchange is disaggregated into import and export for differentiated cost allocation, the heat pump and heating element connections are separated by component and operating mode (SH and DHW), since the electrical power of each serves as an individual de-

cision variable. Additionally, PV curtailment variables are introduced to enable generation reduction under feed-in limitations. The modeling approach further categorizes system elements into four distinct groups: nodes, storages, converters and sources or sinks as described below.

At the DC and AC node, Kirchhoff-type power balance equations enforce conservation among all connected generation, conversion, load, grid and curtailment flows. The storage models comprise a battery energy balance with self-discharge, charge acceptance, softened state of charge (SoC) limits, and a prediction horizon SoC-change variable to prevent myopic decision making, as well as lumped-capacity TES formulations including transmission losses for the SH and DHW tanks with analogously softened temperature bounds. The inverter and charge controller represent converters bounded by nameplate ratings, while the variable-speed heat pump is restricted to operate between outdoor air and supply temperature dependent minimum and maximum compressor speed. Furthermore, the heat pump's thermal output is modeled through a multivariate regression capturing outdoor air temperature, supply temperature, and part-load dependence, complemented by discrete heating element staging, mode-exclusivity and DHW-priority constraints, along with start-up tracking for cycling-cost allocation. Sources and sinks encompass grid exchange limited by connection rating or statutory feed-in cap, curtailment, as well as fixed generation and consumption profiles (exogenous inputs). The detailed system model formulation including underlying equations is provided in Appendix A.1.

2.3. Economic model predictive control

The EMPC approach minimizes a comprehensive cost function J , as defined in Eq. (1), which accounts for both operational expenses and component replacement costs.

$$J = \Delta t \sum_{t=1}^T \begin{bmatrix} c_{P,Grid,im}(t) \\ c_{P,Grid,ex}(t) \\ c_{P,CC,ch} \end{bmatrix}^T \cdot \begin{bmatrix} P_{Grid,im}(t) \\ P_{Grid,ex}(t) \\ P_{CC,ch}(t) \end{bmatrix} + \sum_{t=1}^T \begin{bmatrix} c_{v,HP} \\ c_{e,SoC,Bat} \\ c_{e,\theta,TES,DHW} \\ c_{e,\theta,TES,SH} \end{bmatrix}^T \cdot \begin{bmatrix} v_{HP}(t) \\ \epsilon_{SoC,Bat}(t) \\ \epsilon_{\theta,TES,DHW}(t) \\ \epsilon_{\theta,TES,SH}(t) \end{bmatrix} + c_{\Delta SoC,Bat}(t) \cdot \Delta SoC_{Bat}$$

The cost structure comprises grid interaction costs through electricity purchase prices $c_{P,Grid,im}$ and feed-in tariffs $c_{P,Grid,ex}$ (negative values representing revenue), battery degradation costs $c_{P,CC,ch}$ based on units-of-production depreciation reflecting cycle-induced capacity fade, heat pump start-up costs $c_{v,HP}$ accounting for compressor wear from cycling, and SoC variation costs $c_{\Delta SoC,Bat}$ ensuring appropriate valuation of stored energy beyond the prediction horizon. Slack-variable penalties $c_{e,*}$ are set several orders of magnitude higher than any physical cost¹ to guarantee that violations occur only if no feasible operating point exists and the system states are brought back into the permissible range as quickly as possible. By jointly capturing energy prices, equipment wear, and regulatory constraints within a single optimization objective, this cost structure distinguishes the proposed formulation from conventional MPC approaches. The complete cost coefficient derivation and prioritization conditions are provided in Appendix A.2.

2.4. Rule-based control

To obtain a fair comparison, the reference controllers described in Section 1 are mimicked by a one-step optimization (Fig. 2) whose constraints and cost coefficients are dynamically manipulated by explicit

¹ For instance, the DHW TES temperature violation penalty is set to 10^4 ct/K if grid import prices are on the order of 10^1 ct/kWh.

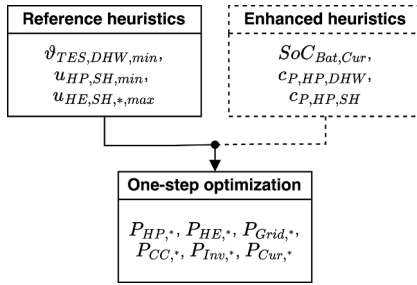


Fig. 2. RBC via heuristics and one-step optimization.

heuristics. This approach enables precise integration of hysteresis behavior and provides ideal PI control without the inevitably associated parameter tuning requirements [21,22,24].

Two RBC variants are implemented within this framework. The reference rule-based control (RRBC) represents conventional practice: the electrical subsystem follows a self-consumption maximization strategy that prioritizes battery charging over grid feed-in and battery discharging over grid import. The thermal subsystem employs a two-point hysteresis controller with fixed maximum compressor speed for DHW preparation–prioritized over space heating–while SH control maintains the heating curve setpoint by varying compressor speed and extends runtime with an upper hysteresis limit. The enhanced rule-based control (ERBC) extends this approach through forecast-based battery management (reserving storage capacity for anticipated PV surplus to minimize curtailment) and surplus-driven heat pump operation that exploits excess generation for thermal storage within elevated temperature bands. Notably, ERBC still applies separate electrical and thermal heuristics without explicit mutual awareness. The detailed algorithms for RRBC and ERBC are presented in Appendices A.3 and A.4, respectively.

3. Results

The impact of energy management strategies on overall system operation—including their behavior under recently introduced regulatory constraints—is evaluated through an annual case study simulation.² This section is organized into four parts presenting a comprehensive evaluation of the proposed EMPC regime against the RRBC and ERBC approaches. First, the case study setup is introduced, detailing the building characteristics, system sizing, demand and generation profiles, as well as regulatory boundary conditions that form the basis for all subsequent analyses. Second, all control strategies are validated and qualitatively compared through the evaluation of high-resolution time series data. This allows for a comprehensive insight into individual operational characteristics of each control strategy. Third, the intertemporal results of each control strategy are quantitatively assessed using aggregated key performance indicators derived from a full-year simulation. Fourth, the findings from both qualitative and quantitative analyses are synthesized and critically examined with regard to the underlying assumptions, simplifications and practical implementation obstacles.

3.1. Case study setup

The analysis is based on a single-family residential building conforming to the German KfW 55 energy efficiency standard, with a reference floor area of 187 m² (parameters taken from the TABULA database [59]), located in Potsdam. This location represents average German climatic conditions and serves as the statutory reference for building energy calculations (DIN/TS 18599-10 [60]).

² The main analysis uses 2025 data; corresponding 2024 and 2023 results are provided in Appendix B.

Table 2
System component ratings.

Component	Nominal rating
Heat pump	5 kW _{th} @ A-7/W35
Heating element	5.5 kW _{AC}
DHW tank	300 L
SH buffer tank	300 L
AC PV	6 kW _p
DC PV	6 kW _p
Inverter	6 kW _{AC}
Charge controller	8 kW _{DC}
Battery	12.5 kWh

System sizing follows established standards if applicable or best practices for both thermal and electrical subsystems. The air-source heat pump with auxiliary electric heater and domestic hot water storage are dimensioned according to VDI 4645 [61] with a bivalent point of −7 °C and heating load calculated per DIN EN 12831-1 [62]. Buffer storage sizing follows the Swiss Wärmepumpen-Systemmodul guideline [63], thereby also satisfying the minimum volume requirements of VDI 4645 [61] and DIN EN 15450 [64]. The roof's maximum south-facing potential for photovoltaic generation is realized with half AC-coupled and half DC-coupled panels. Battery capacity sizing stems from recommendations for usable³ storage capacity limits by Orth, Weniger, and Meissner [65]. The resulting component ratings are summarized in Table 2.

The building is assumed to accommodate four occupants with a total annual consumption of 4000 kWh electricity and 2000 kWh for domestic hot water. Generation and load profiles are based on German Meteorological Service [66] historical weather data for 2025, with PV generation modeled using pvlib [67], space heating demand calculated according to ISO 13790 [68], and household electricity and domestic hot water profiles generated following the Association of German Engineers' methodology for residential reference load profiles [69,70].

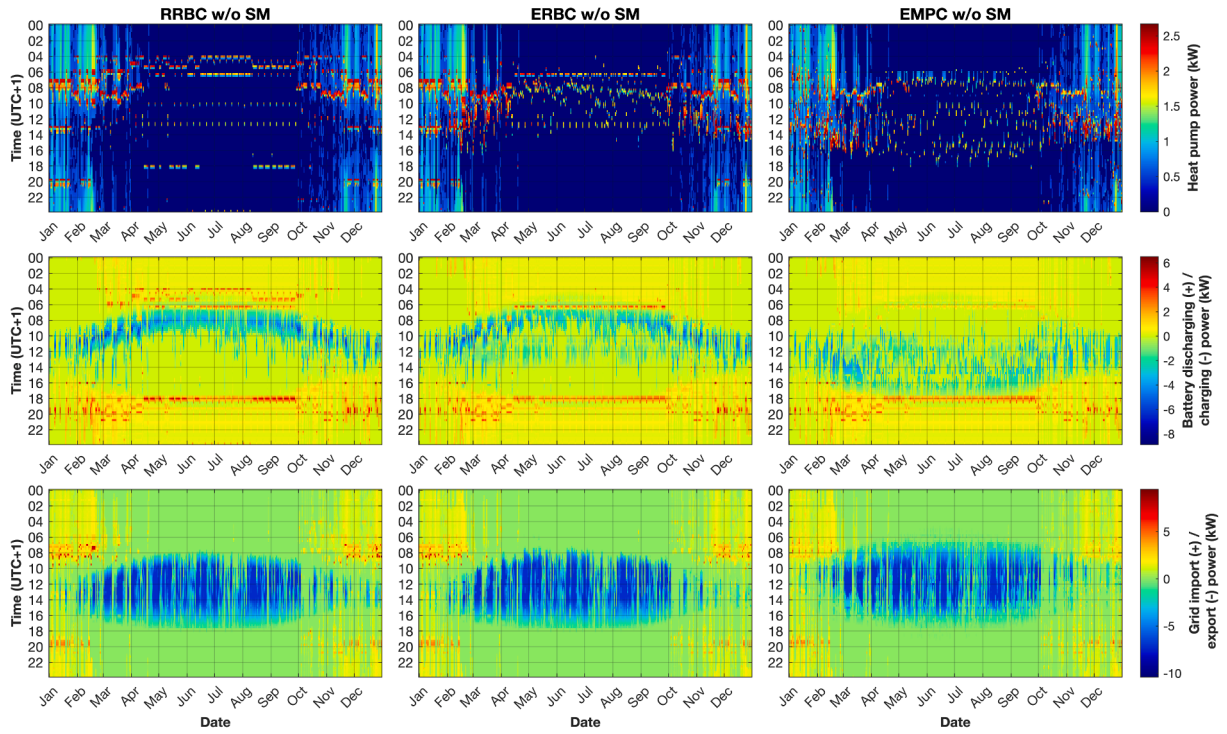
March 1, 2025 is chosen as reference date for electricity tariffs [71] and feed-in compensation rates, leading to the following additional regulatory scenario distinction: without smart meter, the system operates with a static feed-in cap, while with smart meter, the system faces zero compensation during negative day-ahead prices (quarter-hourly historical data for the DE/LU bidding zone obtained from SMARD [72] for 2025). Although not presenting a legal requirement, zero compensation periods are implemented as complete feed-in prohibition for clarity.

All simulations employ perfect foresight and a uniform 15 min time step resolution to strike a balance between temporal accuracy and computational effort. On the one hand, this interval constitutes the coarsest granularity that remains aligned with the quarter-hourly day-ahead spot market and has been proven to provide adequate control responsiveness under real-world disturbances and forecast errors [52,73]. On the other hand, the availability of battery and TES allows for the effective buffering of higher-frequency variability in PV output and electrical or thermal demand, rendering the impact on aggregated results negligible [74,75]. The EMPC is implemented with 24 h prediction horizon (96 time steps), 3 h re-optimization interval (12 time steps) and 5% relative optimality gap to reduce computational burden. For both RBC variants, the underlying one-step optimization problems are solved at each time step using a relative optimality gap of 10^{−9}.

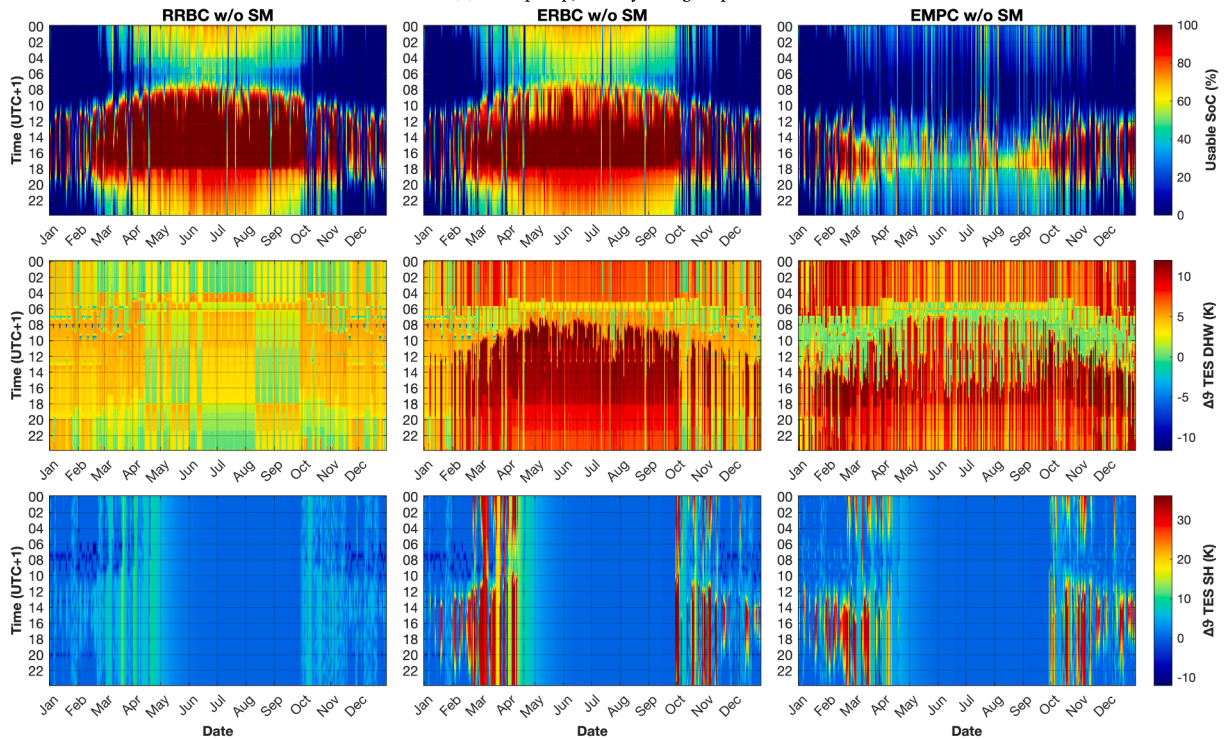
3.2. Qualitative analyses

The annual heatmap visualizations in Figs. 3 and 4 provide comprehensive insight into the temporal system dynamics of system operation under different control regimes, enabling direct comparison of heat pump dispatch, battery utilization, grid interaction, and temperature management strategies across all three approaches. The three main

³ Throughout this paper, usable SoC refers to the nominal range of 10% to 90%.



(a) Heat pump, battery and grid power.



(b) Usable SoC, DHW and SH TES temperature difference relative to minimum required.

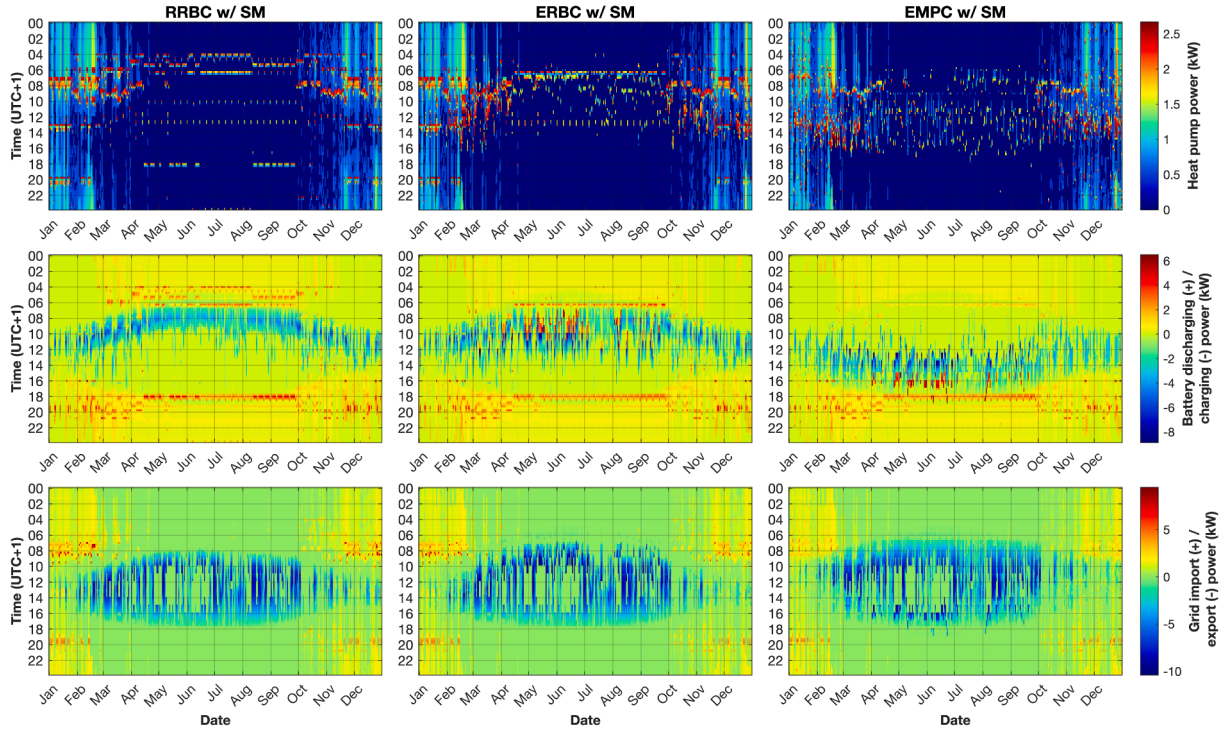
Fig. 3. Annual profiles without smart meter (2025).

characteristic periods—winter, transition, and summer—are analyzed separately in order to reveal seasonal variations.

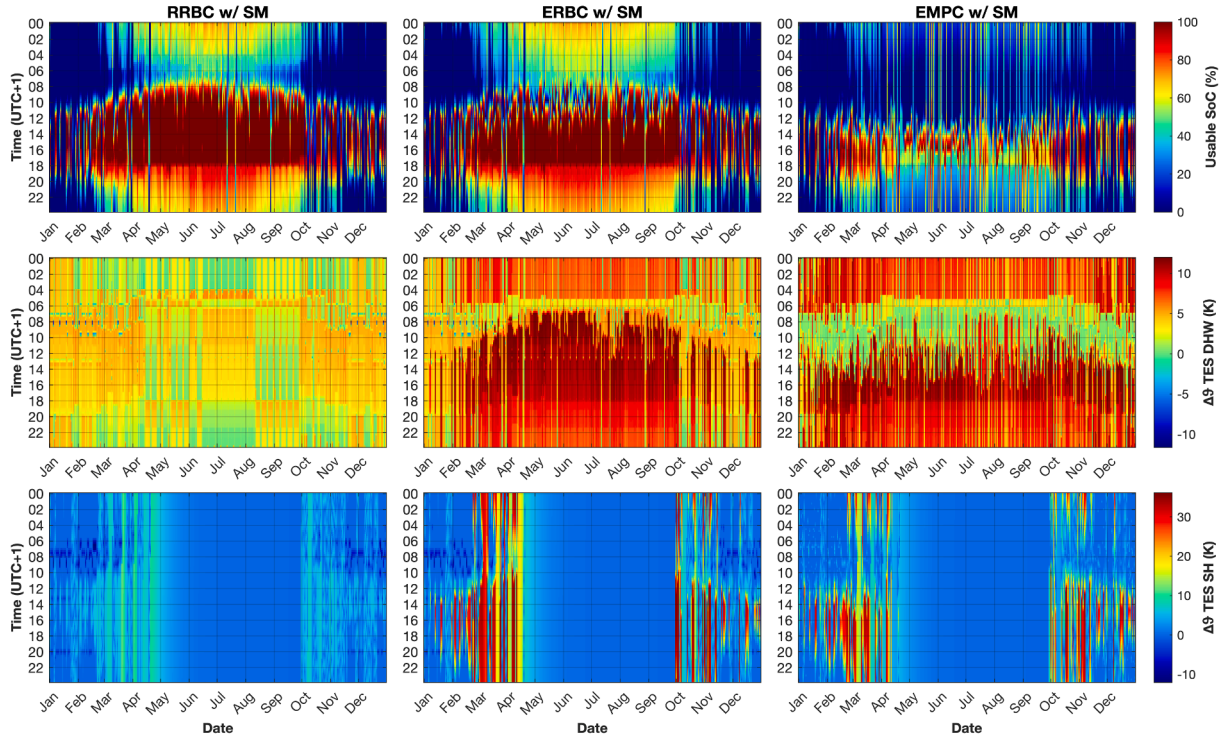
3.2.1. Winter period

The winter period is characterized by predominantly low PV generation coupled with moderate to high space heating demand. Across

all control strategies, several common operational patterns emerge. Heat pump power consumption closely tracks space heating demand throughout this period, with distinct power peaks occurring during DHW TES charging events. Battery utilization remains minimal, with the usable SoC hovering near zero for extended periods. While sporadic afternoon charging brings the battery to full usable capacity,



(a) Heat pump, battery and grid power.



(b) Usable SoC, DHW and SH TES temperature difference relative to minimum required.

Fig. 4. Annual profiles with smart meter (2025).

complete discharge already occurs before midnight. Consequently, continuous grid import dominates the period between sunset and sunrise, with only marginal grid feed-in observed during peak generation hours.

Notable differences between control strategies become apparent in temperature management and operational scheduling. Both RRBC and ERBC briefly violate minimum DHW and SH TES temperature require-

ments, potentially compromising user comfort. In contrast, EMPC maintains all temperature constraints throughout the period. The ERBC and EMPC strategies demonstrate occasional shifting of heat pump operation to midday periods, resulting in moderate overheating of both storage tanks. This behavior is more pronounced in the EMPC implementation, which exploits available PV generation slightly more aggressively for thermal storage.

The absence of significant PV surplus during winter conditions renders the impact of smart meter implementation negligible, with no discernible differences in operational behavior between scenarios with and without dynamic feed-in limitations.

3.2.2. Transition period

The transition period exhibits highly variable day-to-day PV generation ranging from low to high yields, coupled with reduced to moderate space heating demand. All control strategies demonstrate substantially increased battery utilization during this period, with the full usable storage capacity being exploited throughout daily cycles. Grid import, while significantly reduced compared to winter operation, remains concentrated in early morning hours when both PV generation and stored energy are unavailable.

Distinct operational differences emerge in energy management timing and thermal storage strategies. The RRBC and ERBC implementations prioritize early battery charging, capturing PV generation during morning hours to achieve full usable SoC as quickly as possible. Conversely, EMPC delays battery charging until afternoon periods, maintaining lower SoC levels throughout the morning. Grid feed-in patterns follow inverse timing, with RRBC and ERBC exporting later in the day while EMPC feeds surplus power to the grid during morning hours. Both ERBC and EMPC implementations extensively relocate heat pump operation to coincide with peak PV generation, resulting in substantial overheating of both TES. However, EMPC demonstrates more reasonable SH tank temperature increase under mild ambient conditions compared to ERBC. Additionally, progressive improvements in grid independence are evident, with EMPC achieving the lowest import requirements followed by ERBC.

Smart meter integration introduces observable behavioral changes during this period, with occasional battery discharge to grid occurring either during (late) morning hours (ERBC) or afternoon/evening periods (EMPC), strategically timed to avoid zero-compensation.

3.2.3. Summer period

The summer period is characterized by consistently high PV generation and the absence of space heating demand, resulting in operating conditions that mainly differ by battery management and domestic hot water preparation. Across all control strategies, grid imports are almost entirely avoided during this period, while a large share of the PV yield is exported.

The RRBC and ERBC implementations continue to prioritize early battery charging, achieving a full usable SoC by late morning on most days and maintaining the daily average SoC almost exclusively within the upper third of the usable range. In contrast, the EMPC postpones charging to the (late) afternoon, keeping the SoC lower during morning hours. Grid feed-in follows the inverse temporal pattern: RRBC and

ERBC export predominantly later in the day, whereas EMPC feeds surplus power to the grid mainly in the morning. In the thermal subsystem, ERBC and EMPC almost entirely relocate heat pump operation to periods of high solar availability, which leads to pronounced overheating of the DHW storage and makes evening operation largely obsolete. However, EMPC tempers this effect by inducing less extreme and typically later overheating, with daily maximum DHW temperatures not being reached before noon or even later.

Smart meter integration again leads to noticeable changes in the temporal allocation of grid interaction. The midday period of zero feed-in remuneration is clearly reflected in the suppression of grid export, which in turn induces pronounced battery discharge to the grid either in the morning under ERBC or already in the evening under EMPC, strategically anticipating the zero-compensation interval. Under the smart meter scenario, an additional behavioral difference is noteworthy: While EMPC intentionally schedules heat pump operation during zero-compensation intervals, the ERBC's decoupled controller architecture produces a contrasting outcome. As stated above, part of the morning PV surplus (and/or battery discharge) is consumed for thermal storage superheating instead of being fully fed into the grid, resulting in an early satisfaction of the hysteresis temperature threshold that precludes a further heat pump activation within the non-remunerated midday period.

3.3. Quantitative analyses

3.3.1. Energy utilization

Fig. 5 illustrates the annual electrical energy balance, presenting energy supply sources (left bar) and consumption sinks (right bar) for each control strategy. The potentially available PV energy and household electrical load represent fixed variables that remain unchanged across control strategies.

As anticipated, the analysis reveals progressive reductions in grid import, conversion losses, and curtailment with increasing control strategy sophistication. Compared to the RRBC, grid imports fall by 9.7% with the ERBC and 13.1% with EMPC implementation regardless of smart meter integration. Notably, while the RRBC experiences annual curtailment of 287 kWh without and 2789 kWh with smart meter, both the ERBC and EMPC strategies completely eliminate or drastically reduce curtailment by 34.7% and 41.3%, respectively.

The grid feed-in analysis presents more nuanced results. Without smart meter, ERBC reduces grid feed-in by 0.2% compared to the RRBC, whereas EMPC achieves a 1.8% increase. This apparent discrepancy can be attributed to the heat pump and heating element electrical energy consumption patterns: compared to the RRBC, heat pump consumption rises by 2.1% with ERBC but declines 0.8% with EMPC. This finding suggests that while the ERBC effectively utilizes surplus PV generation for thermal storage, it does so less efficiently than the EMPC approach.

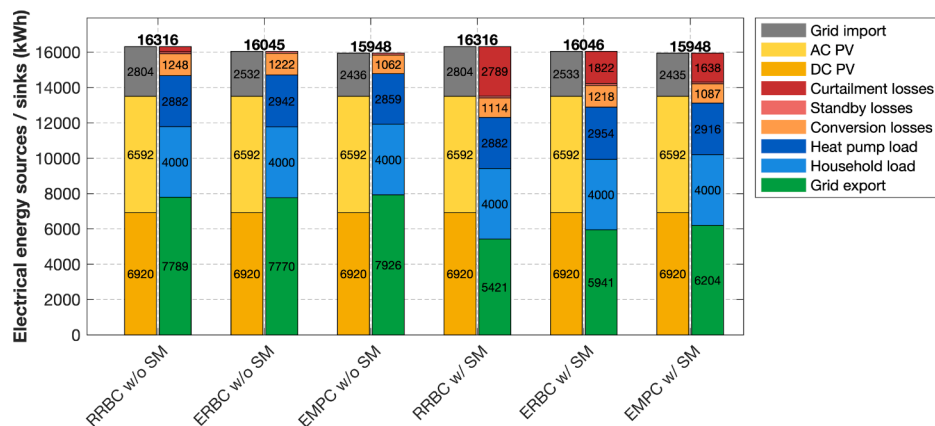


Fig. 5. Annual electrical energy generation and consumption (2025).

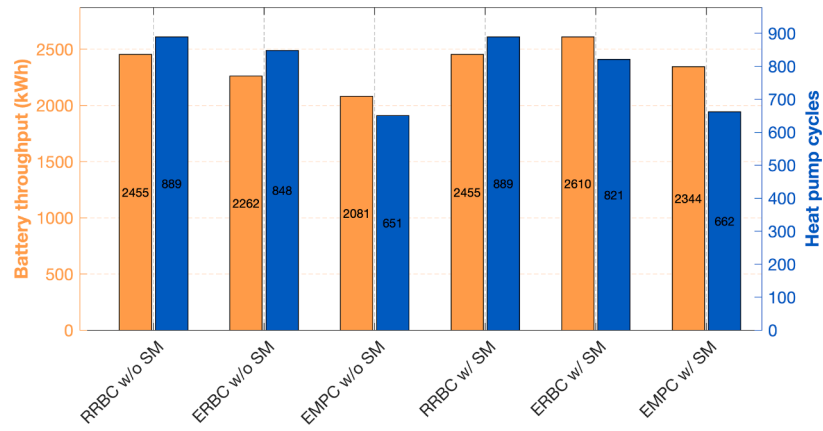


Fig. 6. Annual component degradation parameters (2025).

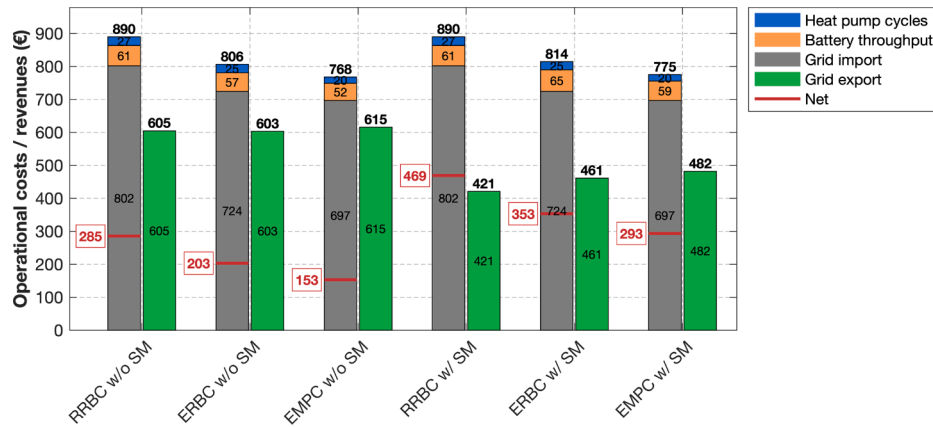


Fig. 7. Costs and revenues (2025).

In contrast, both ERBC and EMPC boost grid export by 9.6% and 14.4% despite growing heat pump demand of 2.5% and 1.2% relative to RRBC with smart meter.

Conversion losses strikingly decline by 2.1% and 14.9% for ERBC and EMPC without smart meter. Regardless of the aforementioned feed-in increase, the EMPC is still capable of maintaining a 2.4% reduction, while the ERBC leads to a 9.3% rise with smart meter.

3.3.2. Component degradation

Fig. 6 presents the component degradation parameters evaluated in this study across the different control strategies. The left y-axis shows annual battery throughput, while the right y-axis presents the heat pump compressor cycles per year.

Both degradation indicators demonstrate substantial improvements with increasing control strategy complexity. Compared to the RRBC baseline, battery throughput decreases by 7.9% with ERBC, 15.2% with EMPC (both without smart meter) and 4.5% with EMPC (with smart meter). With an increase of 6.3%, ERBC constitutes an exception in this regard (with smart meter). Heat pump cycling frequency shows even more dramatic reductions: at least 4.6% with ERBC and 25.5% fewer cycles with EMPC.

The substantial reduction in heat pump cycles achieved by EMPC appears particularly remarkable when considered alongside the marginal differences in energy consumption documented in the previous section. This finding underscores the EMPC's ability to optimize equipment operation patterns while maintaining service quality.

3.3.3. Costs and revenues

Fig. 7 presents the economic analysis encompassing annual costs (left bar), revenues (right bar, generated exclusively from grid export), and

net financial results (line) for each control strategy. Given the assumption of fixed prices throughout the simulation period, all costs and revenues maintain direct proportionality to the physical parameters presented in the preceding sections.

Grid electricity import constitutes the predominant cost component across all control strategies, accounting for roughly 90% of total operational costs at a minimum. Battery throughput costs represent the second-largest expense category, followed by heat pump cycling costs.

The economic analysis reveals that grid export revenues prove insufficient to offset operational costs under all control strategies, resulting in negative net financial outcomes in all cases. Nevertheless, significant savings are achievable through advanced control implementation. The ERBC shows a reduction of 28.8% and 24.7% compared to the reference baseline, while the EMPC implementation nearly doubles that to 46.4% and 37.5% without and with smart meter, respectively, demonstrating the substantial economic benefits of sophisticated control strategies in building MESs.

3.4. Discussion

3.4.1. Interpretation of results

The comprehensive analysis demonstrates that while heuristic extensions to the RRBC can improve overall operating performance, EMPC exhibits clear superiority across all evaluated metrics. Through control strategy modifications alone, the system achieves reduced grid consumption, increased feed-in revenues, and extended operational lifetimes for both battery and heat pump.

Fig. 8 links short-term operational behavior with long-term performance metrics. While the composition of household load coverage remains effectively constant across control strategies due to its non-

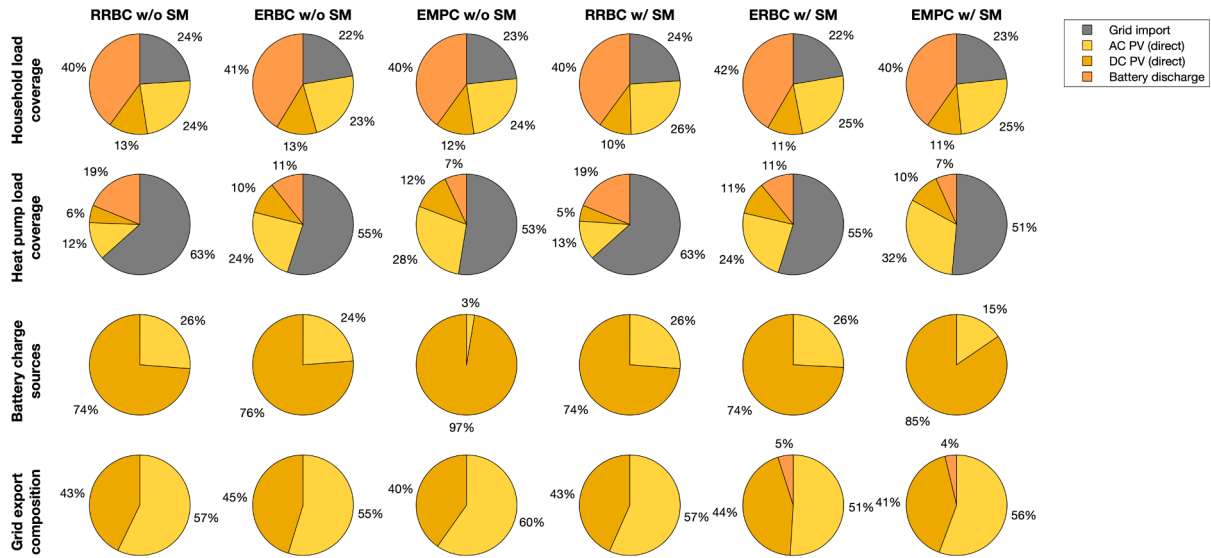


Fig. 8. Shares of household and heat pump load coverage, battery charging sources and grid export composition (2025).

shiftable nature, striking differences emerge in heat pump load coverage and battery charging sources.

Regarding heat pump load coverage, both EMPC and ERBC substantially increase the direct PV consumption share, thereby reducing grid dependency through effective utilization of TES, which not only minimizes battery throughput but also effectively extends total available storage capacity. The EMPC achieves this objective with considerably greater precision, leveraging its knowledge of part-load, outdoor-, and TES temperature-dependent coefficient of performance (COP), combined with accurate future consumption profiles. In contrast, under the ERBC's unnecessary overheating, feed-in revenues are reduced without smart meter integration. Nevertheless, this approach remains economically rational, especially during winter and transition period: although overheating lowers the heat pump's efficiency, the resulting increase in energy consumption costs less than the alternative of feeding excess PV into the grid and later purchasing electricity at higher rates.

The battery charging analysis reveals particularly striking differences. To minimize AC to DC conversion losses (and vice versa) through the inverter, EMPC only stores DC PV generation in the battery when absolutely necessary (either for load supply or curtailment reduction), while directing AC PV power to TES via heat pump operation. This behavior simultaneously reduces battery degradation by minimizing throughput. Although SoC-dependent calendar aging is not explicitly accounted for in the objective function, this operating regime may be relevant for battery lifetime, as calendar aging is commonly aggravated at high SoC [76].

The primary mechanisms for reduced heat pump cycling with EMPC include immediate switching between DHW and SH modes without operational downtime, combined with the previously discussed strategic TES superheating. Consequently, the EMPC's energy distribution follows no static sequence but is dynamically optimized for each time step based on prevailing conditions and future predictions.

While the presented results show qualitative agreement with findings from related studies, quantitative comparisons, as mentioned earlier, prove challenging due to fundamental differences in system architectures and control approaches. Tjaden et al. [25] reported slightly smaller savings between RRBC and ERBC implementations; however, their analysis was constrained by several methodological limitations: the use of standardized SG Ready interfaces precluded direct power modulation with no optimization-based control strategy evaluated for heat pump operation. Fischer et al. [21] likewise confirmed clear EMPC advantages over ERBC, yet their framework differs substantially from

the present work in several key aspects: battery storage was excluded from the system configuration, TES temperatures were estimated a priori rather than optimized simultaneously with other control variables, and their quadratic programming formulation necessitated post-processing to obtain implementable control actions. The most comparable study in terms of control strategy scope (RRBC and EMPC) as well as physical system configuration (PV, battery, air-source heat pump, SH and DHW TES) is found in Kuboth et al. [42], who achieved 12% operational cost savings through nonlinear EMPC that additionally exploits building thermal mass via zone temperature optimization. Their framework differs from the presented work in several aspects: it neglects part-load variations in heat pump COP, relies on a particle swarm solver, and decomposes the problem into successive sub-optimizations interleaved with heuristic post-processing, which does not inherently guarantee global optimality. In contrast, the present study integrates all major system components—battery storage, variable-speed heat pump, and TES—within a unified MILP framework that enables simultaneous optimization of electrical and thermal domains without requiring post-processing.

Beyond methodological differences, the present study distinguishes itself by explicitly considering two regulatory scenarios for PV feed-in management. While related studies typically assume either unrestricted feed-in or static limitations, this work evaluates control strategy performance under both static power caps and dynamic zero-compensation periods. This distinction proves particularly relevant for the EMPC approach, which demonstrates superior capability in responding to time-varying feed-in conditions, as evidenced by the strategic battery charge/discharge patterns observed in the summer period analysis. Existing comparative studies do not address this regulatory dimension, limiting their applicability to evolving market conditions.

3.4.2. Limitations and outlook

Despite the rigorous validation efforts employed in this case study, comprehensive sensitivity analyses remain necessary for generalizing the results to broader applications. Several assumptions and simplifications impact both the results and real-world implementation feasibility.

The key assumptions affecting the analysis include perfect foresight capabilities. While various studies suggest prediction errors may not critically impact performance [21,77], the absence of explicit uncertainty modeling remains a limitation. Nevertheless, the deterministic formulation establishes a necessary baseline for isolating structural differences between control strategies before introducing additional sources of variability. Importantly, simple analyses based on fixed perturbations of individual forecast inputs would not fully capture the

correlated error structures inherent to simultaneous weather, generation, and demand forecasting [78,79]. Notwithstanding, the proposed EMPC framework has been successfully operated in a real-world installation without perfect foresight [73], substantiating its applicability under realistic information conditions. The assumption of ideal PI control may overestimate RBC performance capabilities in practice. Additionally, the degradation cost assignment methodology lacks well-established theoretical foundations and empirical long-term studies for battery and heat pump wear in unit commitment and dispatch optimization contexts [24,80].

The modeling simplifications include the omission of heat pump start-up losses, which are expected to have only minor effects given the long operational periods (even with RBC) compared to the relevant time constants [81]. The lumped-capacitance approach, however, may overestimate potential benefits: Schütz et al. [50] reported a 4% annual cost increase for SH heat pump and TES with variable grid import when stratified models were used. This effect, however, is strongly influenced by sensor placement [21] and is not necessarily unidirectional: especially for DHW storage, the uniform temperature constraint of the lumped formulation can restrict the admissible operating range compared to a stratified model that only requires tapping temperature in the upper extraction zone, potentially underestimating usable capacity [6]. Furthermore, the quarter-hourly resolution neglects conversion losses and fluctuations resulting from highly volatile generation or consumption patterns occurring within the sampling period.

Beyond these assumptions, the SH demand is treated as an exogenous time series, restricting thermal load-shifting in the EMPC to the TES. Embedding indoor temperature as a dynamic state variable and exploiting the building mass could provide additional flexibility. However, doing so typically requires zone-level sensing (e.g., temperature and heat flow measurements) as well as a calibrated thermal model [35,44], which are beyond the low-instrumentation EMPC framework for hydronic central heating systems considered in this work. The same sensing and modeling constraints are applied consistently across all compared control strategies, ensuring that observed differences arise from the control formulations rather than from differences in available information.

Still, barriers to broad industry adoption remain significant, particularly the absence of standardized (part-load) test data and communication interfaces. This limitation proves especially problematic for heat pump systems, where proprietary protocols and inconsistent data formats complicate the implementation of advanced control strategies across heterogeneous equipment portfolios. While the open-source EE-Bus protocol seeks to standardize communication between building MES components, direct implementation of the EMPC power control is not (yet) supported by its predefined use case structure [82]. However, equivalent functionality can already be achieved by combining use cases such as “Power Limitation” or “Self Consumption Optimization” with strategic temperature setpoint manipulation. The authors have validated this approach in practice, utilizing the standard Modbus interface of a commercially available heat pump to access the required parameters [73].

Several promising avenues for future research emerge from this work. First, extending the control framework to incorporate day-ahead market-based dynamic pricing could unlock economic benefits for end-users through cost reduction and/or revenue generation via grid services. Second, environmental impact assessments featuring marginal emission factors would quantify the carbon reduction potential of advanced control strategies. Third, the integration of electric mobility with both vehicle-to-home and vehicle-to-grid capabilities represents a natural extension of the MES concept. Fourth, incorporating the refrigerant cycle reversal capability as a third heat pump operating mode—combined with sensible or latent cold storage or indoor temperature control as discussed above—would enable the EMPC to exploit analogous load-shifting potentials in summer cooling scenarios. Fifth, addressing prediction uncertainty through probabilistic forecasting and robust or stochastic optimization approaches could enhance practical performance, al-

though the increased computational effort of the latter compared to the proposed deterministic framework may impair its real-time capability [7,83], which currently represents a key strength. Finally, reformulating the optimization as a mixed-integer quadratically constrained program would support both the integration of stratified TES models and optimal component sizing, capturing additional operational efficiency and design improvements.

4. Conclusion

This study presented a comprehensive comparison between RBC and EMPC strategies for building MESs, demonstrating the substantial benefits of advanced optimization-based approaches. The primary contribution of this research is the development of a holistic MILP-based EMPC framework that simultaneously integrates electricity and heating sectors while maintaining computational efficiency for real-time implementation. The framework was evaluated against state-of-the-art RBC strategies using consistent performance metrics, addressing a significant gap in the literature. Additionally, by evaluating control strategies under both static feed-in power caps and dynamic zero-compensation periods, this work provides timely insights into the implications of recently enacted regulatory changes for building energy management. Furthermore, the detailed component models incorporating part-load and temperature-dependent heat pump performance, battery charge acceptance, and comprehensive degradation cost allocation improve the representation of realistic MES operation. The key findings reveal that while enhanced rule-based control with surplus-driven heat pump operation can improve system performance, EMPC achieves superior results across all evaluated metrics. Specifically, the proposed control strategy reduces net annual costs by 37.5%, grid imports by 13.1%, battery throughput by up to 15.2%, heat pump cycling by 25.5%, and PV curtailment by 41.3% compared to the reference approach. Overall, these results confirm that EMPC’s cross-domain application under realistic system configurations and current regulatory conditions yields substantial performance improvements that justify the additional implementation complexity. The proposed framework therefore provides a promising basis for the operation of future residential MESs under increasingly dynamic electricity markets and evolving regulatory frameworks.

CRedit authorship contribution statement

Niklas Kießling: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Daniel Wolf:** Writing – review & editing, Supervision, Conceptualization; **Fontina Petrakopoulou:** Writing – review & editing, Supervision.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude and ChatGPT in order to improve the language of the manuscript. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support by the Federal Ministry for Economic Affairs and Climate Action (BMWK), promotional reference 03EGB0025.

Data availability

Data will be made available on request.

Appendix A. Methodology

A.1. Detailed system model formulation

All conversion efficiencies, losses and operational constraints are concentrated in the connected component models so that Kirchhoff-type balance equations can be imposed at the DC and AC electrical nodes as well as at both TES. Throughout the formulation, actual decision variables of the optimization problem are distinguished using bold font notation, while all other variables represent either linear combinations of these or fixed predetermined inputs, outputs, and parameters (see [84] for fundamentals of model predictive control). For readers interested in the fundamentals of MILP in unit commitment and dispatch optimization, Morales-España, Latorre, and Ramos [85] or Knueven, Ostrowski, and Watson [86] provide essential information on tightly formulating start-up and shut-down power trajectories of thermal units.

A.1.1. Nodes

DC node. The DC node serves as the central junction for direct current power flows within the system. Power conservation is enforced by Eq. (A.1) encompassing the physical powers of DC PV, charge controller, inverter, and standby losses. To accurately account for conversion losses in both charging and discharging directions, the charge controller and inverter power flows are explicitly separated. Furthermore, the formulation includes a curtailment power variable that enables the limitation of DC PV generation when necessary due to inverter capacity constraints or grid export limitations.

$$0 = P_{PV,DC}(t) + P_{CC,dch}(t) + P_{Inv,AC/DC}(t) - P_{Stby}(t) - P_{CC,ch}(t) - P_{Inv,DC/AC}(t) - P_{Cur,DC}(t) \quad (A.1)$$

AC node. The AC node balance, formulated in Eq. (A.2), captures the alternating current power flows of PV, inverter with directional efficiency factors ($\eta_{Inv,DC/AC}$ for DC to AC conversion and $\eta_{Inv,AC/DC}$ for AC to DC conversion), grid interaction, household electrical loads, heat pump and heating element (latter two further distinguished by operating mode). Grid power is disaggregated into import and export components to enable differentiated cost allocation within the optimization framework. Similar to the DC node, AC PV curtailment capability is also incorporated.

$$0 = P_{PV,AC}(t) + \eta_{Inv,DC/AC} \cdot P_{Inv,DC/AC}(t) + P_{Grid,im}(t) - P_{Load}(t) - \frac{P_{Inv,AC/DC}(t)}{\eta_{Inv,AC/DC}} - P_{Grid,ex}(t) - P_{Cur,AC}(t) - P_{HP,SH}(t) - P_{HE,SH}(t) - P_{HP,DHW}(t) - P_{HE,DHW}(t) \quad (A.2)$$

A.1.2. Storages

Battery. The battery energy balance, defined by Eq. (A.3), relates the change in stored energy to the charge controller power flows and self-discharge losses. The formulation incorporates the nominal battery capacity $E_{Bat,nom}$, self-discharge coefficient $b_{Bat,SD}$, and directional efficiency parameters for the charge controller ($\eta_{CC,ch}$ for charging and $\eta_{CC,dch}$ for discharging).

$$E_{Bat,nom} \cdot \frac{SoC_{Bat}(t) - SoC_{Bat}(t-1)}{\Delta t} = \eta_{CC,ch} \cdot P_{CC,ch}(t) - \frac{P_{CC,dch}(t)}{\eta_{CC,dch}} - b_{Bat,SD} \cdot E_{Bat,nom} \cdot SoC_{Bat}(t-1) \quad (A.3)$$

SoC constraints bound the battery operation within feasible limits through constant minimum and maximum values. To ensure mathematical feasibility and maintain system operation under all circumstances,

the constraints are softened using 1-norm formulation with highly penalized slack variables $\epsilon_{SoC,Bat} > 0$, which allows temporary constraint violations only when absolutely necessary.

$$SoC_{Bat}(t) - \epsilon_{SoC,Bat}(t) \leq SoC_{Bat,max} \quad (A.4)$$

$$SoC_{Bat}(t) + \epsilon_{SoC,Bat}(t) \geq SoC_{Bat,min} \quad (A.5)$$

An additional decision variable, introduced in Eq. (A.6), captures the change in battery SoC (net discharge) over the prediction horizon T . This formulation, similar to terminal cost concepts in model predictive control, ensures that energy stored beyond the prediction horizon is appropriately valued in the cost function, preventing myopic optimization decisions.

$$\Delta SoC_{Bat} = SoC_{Bat}(0) - SoC_{Bat}(T) \quad (A.6)$$

Thermal energy storages. The lumped-capacity models of the SH and DHW tank follow similar mathematical structures. The SH TES balance, expressed in Eq. (A.7), accounts for the change in stored thermal energy through heat pump and heating element contributions, thermal demand satisfaction, and transmission losses to the installation environment. The formulation incorporates the TES heat capacity $C_{TES,SH}$, thermal transmittance $U_{TES,SH}$, surface area $A_{TES,SH}$, heating element efficiency η_{HE} , and installation room temperature ϑ_{Inst} . The DHW TES follows an identical formulation with appropriate index substitutions.

$$C_{TES,SH} \cdot \frac{\vartheta_{TES,SH}(t) - \vartheta_{TES,SH}(t-1)}{\Delta t} = \dot{Q}_{HP,SH}(t) + \eta_{HE,SH} \cdot P_{HE,SH}(t) - \dot{Q}_{SH}(t) - U_{TES,SH} \cdot A_{TES,SH} \cdot (\vartheta_{TES,SH}(t-1) - \vartheta_{Inst}) \quad (A.7)$$

Temperature constraints, defined by Inequalities (A.8) and (A.9), establish operational bounds for the TES. The minimum temperature requirement is determined by heating curve methodology [87] using mixed outdoor temperature [88] for SH applications, while DHW follows daily and weekly scheduling patterns. Maximum temperature limits prevent limescale formation and scalding hazards. Again, constraint softening through 1-norm slack variables ensures mathematical feasibility under all operating conditions.

$$\vartheta_{TES,SH}(t) - \epsilon_{\vartheta,TES,SH}(t) \leq \vartheta_{TES,SH,max} \quad (A.8)$$

$$\vartheta_{TES,SH}(t) + \epsilon_{\vartheta,TES,SH}(t) \geq \vartheta_{TES,SH,min}(t) \quad (A.9)$$

A.1.3. Converters

Inverter. The inverter power flows (on the DC node side) in both DC to AC and AC to DC conversion directions are limited by zero and their respective nameplate maxima.

$$0 \leq P_{Inv,DC/AC}(t) \leq P_{Inv,DC/AC,max} \quad (A.10)$$

$$0 \leq P_{Inv,AC/DC}(t) \leq P_{Inv,AC/DC,max} \quad (A.11)$$

Charge controller. Regarding the charge controller, in addition to static bounds in both directions reflecting the installed nameplate power ratings, charging capability is reduced linearly (utilizing coefficient $b_{Bat,CA}$) with increasing SoC level to model charge acceptance of cell batteries [89,90].

$$0 \leq P_{CC,dch}(t) \leq P_{CC,dch,max} \quad (A.12)$$

$$0 \leq P_{CC,ch}(t) \leq P_{CC,ch,max} \quad (A.13)$$

$$P_{CC,ch}(t) \leq b_{Bat,CA} \cdot (1 - SoC_{Bat}(t)) \quad (A.14)$$

Heat pump with heating element. The heat pump modeling approach presented in Eq. (A.15) captures the complex condenser heat flow dependence on outdoor air temperature (ϑ_e), supply temperature ($\vartheta_{HP,*}$), and part-load operation through multivariate regression, where outdoor air temperature enters implicitly via pre-computed time-varying coefficients $b_{HP,\dot{Q},*}(t)$ that aggregate nonlinear interaction terms.

$$\dot{Q}_{HP,SH}(t) = b_{HP,\dot{Q},a}(t) \cdot u_{HP,SH}(t) + b_{HP,\dot{Q},p}(t) \cdot P_{HP,SH}(t) + b_{HP,\dot{Q},\vartheta}(t) \cdot \vartheta_{HP,SH}(t) \quad (A.15)$$

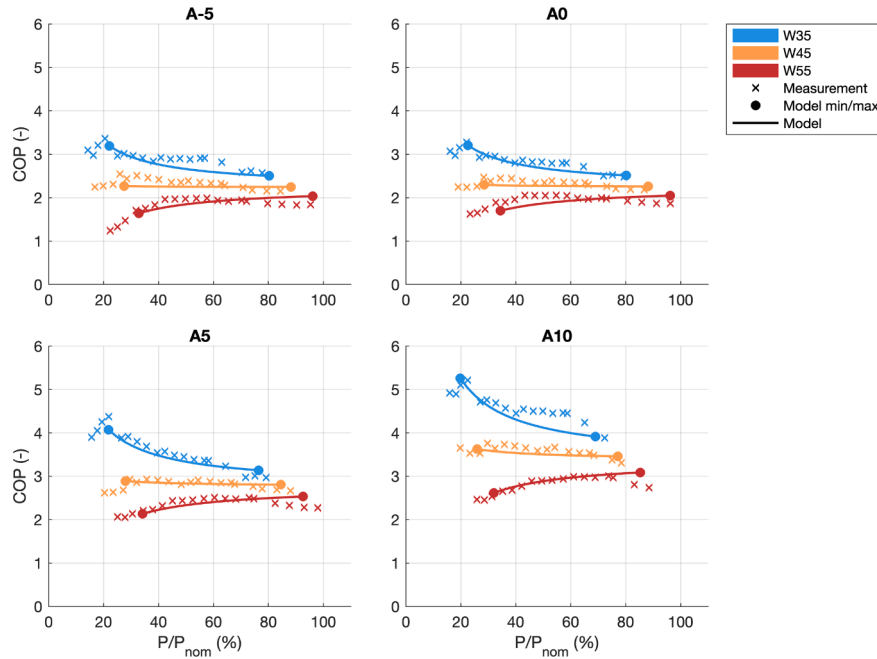


Fig. A.1. Air-to-water heat pump performance map (defrosting loss included).

Curve fitting utilizes either manufacturer data per EN 14511-2 [91] or thermodynamic simulations with defrosting loss considerations [92]. While supply temperature and electrical power appear explicitly as linear terms, outdoor air temperature enters implicitly through nonlinear interaction terms within the aggregated coefficients (computed for each time step's specific outdoor temperature).

$$\begin{aligned} \dot{Q}_{HP,SH}(t) = & \begin{bmatrix} b_{HP,\dot{Q},u,0} & b_{HP,\dot{Q},u,\vartheta_e} & b_{HP,\dot{Q},u,\vartheta_e^2} \end{bmatrix} \cdot \begin{bmatrix} u_{HP,SH}(t) \\ \vartheta_e(t) \cdot u_{HP,SH}(t) \\ \vartheta_e^2(t) \cdot u_{HP,SH}(t) \end{bmatrix} \\ & + \begin{bmatrix} b_{HP,\dot{Q},P,0} & b_{HP,\dot{Q},P,\vartheta_e} & b_{HP,\dot{Q},P,\vartheta_e^2} \end{bmatrix} \cdot \begin{bmatrix} P_{HP,SH}(t) \\ \vartheta_e(t) \cdot P_{HP,SH}(t) \\ \vartheta_e^2(t) \cdot P_{HP,SH}(t) \end{bmatrix} \\ & + \begin{bmatrix} b_{HP,\dot{Q},\vartheta,0} & b_{HP,\dot{Q},\vartheta,\vartheta_e} & b_{HP,\dot{Q},\vartheta,\vartheta_e^2} \end{bmatrix} \cdot \begin{bmatrix} \vartheta_{HP,SH}(t) \\ \vartheta_e(t) \cdot \vartheta_{HP,SH}(t) \\ \vartheta_e^2(t) \cdot \vartheta_{HP,SH}(t) \end{bmatrix} \end{aligned} \quad (A.16)$$

The regression coefficients are obtained by minimizing the mean squared error of the COP rather than thermal output directly, as the latter would disproportionately weight high-capacity operating points due to larger absolute deviations. This approach ensures consistent model accuracy across the entire modulation range—a critical requirement especially for air-to-water heat pumps given their substantial part-load time share in building applications. An exemplary result of the parameter identification procedure is illustrated in Fig. A.1, demonstrating good agreement between manufacturer data and the fitted performance map across the operating envelope.

Compressor operation is bounded within the speed range between optimal and maximum power [21]. This approach circumvents the issue of sharply declining efficiency at very low speeds and ensures monotonicity of the COP function with respect to power. Temperature-dependent bounds (outdoor air and supply) are implemented using the same interaction term principle.

$$\begin{aligned} P_{HP,SH}(t) \geq & b_{HP,P,min,u}(t) \cdot u_{HP,SH}(t) \\ & + b_{HP,P,min,\vartheta}(t) \cdot \vartheta_{HP,SH}(t) \end{aligned} \quad (A.17)$$

$$\begin{aligned} P_{HP,SH}(t) \leq & b_{HP,P,max,u}(t) \cdot u_{HP,SH}(t) \\ & + b_{HP,P,max,\vartheta}(t) \cdot \vartheta_{HP,SH}(t) \end{aligned} \quad (A.18)$$

The heat pump supply temperature formulation, defined by Inequalities (A.19) to (A.21), employs Big-M method [93] to serve multiple purposes: ensuring supply temperature equals zero when the heat pump is inactive (as required by the heat output equation), setting supply temperature equal to the TES temperature plus a constant spread (common charging pump control [94–96]) when active, and restricting heat pump operation to (outdoor-air temperature dependent) feasible supply temperature ranges.

$$\vartheta_{HP,SH,min}(t) \cdot u_{HP,SH}(t) \leq \vartheta_{HP,SH}(t) \leq \vartheta_{HP,SH,max}(t) \cdot u_{HP,SH}(t) \quad (A.19)$$

$$\begin{aligned} \vartheta_{HP,SH}(t) \leq & \vartheta_{TES,SH}(t) + \Delta\vartheta_{HP,SH} \\ & + M \cdot (1 - u_{HP,SH}(t)) \end{aligned} \quad (A.20)$$

$$\begin{aligned} \vartheta_{HP,SH}(t) \geq & \vartheta_{TES,SH}(t) + \Delta\vartheta_{HP,SH} \\ & - M \cdot (1 - u_{HP,SH}(t)) \end{aligned} \quad (A.21)$$

The heating element is implemented using two binary variables, whose possible combinations yield three discrete equidistant power levels:

$$P_{HE,SH}(t) = \frac{P_{HE,max}}{3} \cdot (u_{HE,SH,1}(t) + 2 \cdot u_{HE,SH,2}(t)) \quad (A.22)$$

Mutual-exclusivity constraints (Eqs. (A.23)–(A.25) utilizing the example of $u_{HE,SH,*}$) prevent simultaneous activity of SH and DHW mode.

$$u_{HP,SH}(t) + u_{HP,DHW}(t) \leq 1 \quad (A.23)$$

$$u_{HE,SH,1}(t) \leq u_{HP,SH}(t) \quad (A.24)$$

$$u_{HE,SH,2}(t) \leq u_{HP,SH}(t) \quad (A.25)$$

DHW preparation is prioritized over SH through Inequality (A.26), ensuring minimum temperature requirements are satisfied before SH provisioning.

$$M \cdot (u_{HP,SH}(t) - 1) \leq \vartheta_{TES,DHW}(t) - \vartheta_{TES,DHW,min}(t) \quad (A.26)$$

Additionally, start-up events are tracked by the binary variable v_{HP} (see Morales-España, Latorre, and Ramos [85] for a detailed formulation), enabling cycling cost allocation.

A.1.4. Sources and sinks

Grid. Grid exchange is limited by the maximum grid connection power rating for import and—in absence of smart meter—the statutory feed-in cap (fixed percentage of nominal PV capacity) for export, which usually lies below the physical power rating.

$$0 \leq P_{Grid,im}(t) \leq P_{Grid,im,max}(t) \quad (A.27)$$

$$0 \leq P_{Grid,ex}(t) \leq P_{Grid,ex,max}(t) \quad (A.28)$$

Curtailment. PV curtailment variables are lower-bounded by zero and upper-bounded by the corresponding maximum available generation at each time step, allowing for voluntary power reduction when economically beneficial or operationally necessary.

$$0 \leq P_{Cur,DC}(t) \leq P_{PV,DC}(t) \quad (A.29)$$

$$0 \leq P_{Cur,AC}(t) \leq P_{PV,AC}(t) \quad (A.30)$$

Generation and consumption. Generation ($P_{PV,DC}$, $P_{PV,AC}$) and consumption time series (P_{Load} , $\dot{Q}_{SH}$, $\dot{Q}_{DHW}$, P_{Siby}) are treated as exogenous inputs, generated outside the optimization framework.

A.2. Cost coefficient derivation

A.2.1. Battery throughput

Battery degradation costs are addressed through the units-of-production depreciation method that reflects cycle-induced capacity fade leading to end-of-life conditions, typically defined as 80% of initial capacity retention [97,98]. Consequently, the charging power penalty, calculated according to Eq. (A.31), distributes replacement costs across the expected lifetime based on energy throughput.

$$c_{P,CC,ch} = \frac{\eta_{CC,ch} \cdot \text{battery replacement cost}}{\text{usable battery capacity} \cdot \text{expected battery cycle life}} \quad (A.31)$$

A.2.2. Heat pump cycling

Heat pump compressor wear, primarily influenced by cycling frequency rather than operating hours [24,99,100], is incorporated through start-up cost allocation following Eq. (A.32), incentivizing longer operational periods.

$$c_{v,HP} = \frac{\text{compressor replacement cost}}{\text{expected compressor on/off cycles before breakdown}} \quad (A.32)$$

A.2.3. Battery SoC

The battery SoC change cost $c_{\Delta SoC,Bat}$ ensure appropriate valuation of energy stored beyond the prediction horizon, encouraging charging from renewable sources over feed-in while favoring discharge over grid import by meeting the conditions below:

$$\frac{c_{P,CC,ch} - \frac{c_{P,Grid,ex}}{\eta_{Inv,AC/DC}}}{\eta_{CC,ch}} < \frac{c_{\Delta SoC,Bat}}{E_{Bat,nom}} < \eta_{CC,dch} \cdot \eta_{Inv,DC/AC} \cdot c_{P,Grid,im} \quad (A.33)$$

A.3. Reference heuristics

Reference heuristics only affect the thermal subsystem and comprise three Algorithms: Algorithms 1 and 2 ensure that the heat pump operates within a hysteresis band (upper $\vartheta_{TES,*}$ and lower $\underline{\vartheta}_{TES,*}$ threshold values indicated by over- and underline) in both modes. If the TES temperature falls $\underline{\Delta\vartheta}_{TES,*}$ below the lower threshold or fails to reach latter after $d_{TES,*max}$ ($d_{TES,*}$ representing mode specific operation below $\underline{\vartheta}_{TES,*}$), the back-up heating element is unblocked. The differences between DHW and SH provisioning obtained through Algorithm 3 are as follows: In DHW mode (prioritized), the heat pump runs at maximum power until the upper threshold value is reached. In SH mode, the heat pump continues to run at minimum power after the lower threshold value is exceeded, until the upper threshold value is reached.

Algorithm 1 Demand driven domestic hot water request.

Input: $\vartheta_{TES,DHW}$, $\underline{\vartheta}_{TES,DHW}$, $\underline{\Delta\vartheta}_{TES,DHW}$, $\vartheta_{TES,DHW}$

Output: R_{DHW}

```

1: static  $R_{DHW} \leftarrow 0$ 
2: if  $\vartheta_{TES,DHW} \leq \underline{\vartheta}_{TES,DHW} - \underline{\Delta\vartheta}_{TES,DHW}$  or  $d_{TES,DHW} \geq d_{TES,DHW,max}$  then
3:    $R_{DHW} \leftarrow 2$ 
4: else if  $\vartheta_{TES,DHW} \leq \underline{\vartheta}_{TES,DHW}$  and  $R_{DHW} \neq 2$  then
5:    $R_{DHW} \leftarrow 1$ 
6: else if  $\vartheta_{TES,DHW} \geq \underline{\vartheta}_{TES,DHW}$  then
7:    $R_{DHW} \leftarrow 0$ 
8: end if

```

Algorithm 2 Demand driven space heat request.

Input: $\vartheta_{TES,SH}$, $\underline{\vartheta}_{TES,SH}$, $\underline{\Delta\vartheta}_{TES,SH}$, $\vartheta_{TES,SH}$

Output: R_{SH}

```

1: static  $R_{SH} \leftarrow 0$ 
2: if  $\vartheta_{TES,SH} \leq \underline{\vartheta}_{TES,SH} - \underline{\Delta\vartheta}_{TES,SH}$  or  $d_{TES,SH} \geq d_{TES,SH,max}$  then
3:    $R_{SH} \leftarrow 2$ 
4: else if  $(\vartheta_{TES,SH} \leq \underline{\vartheta}_{TES,SH} \text{ and } R_{SH} \neq 2)$  or  $(R_{SH} = 2 \text{ and } R_{DHW} \neq 0)$  then
5:    $R_{SH} \leftarrow 1$ 
6: else if  $\vartheta_{TES,SH} \geq \underline{\vartheta}_{TES,SH}$  then
7:    $R_{SH} \leftarrow 0$ 
8: end if

```

Algorithm 3 Reference one-step optimization problem adaption.

Input: $\vartheta_{TES,DHW}$, R_{DHW} , R_{SH}

Output: $\vartheta_{TES,DHW,min}$, $u_{HP,SH,min}$, $u_{HE,*}, u_{HE,*max}$

```

1: if  $R_{DHW} = 2$  then
2:    $\vartheta_{TES,DHW,min} \leftarrow \underline{\vartheta}_{TES,DHW}$ 
3:    $u_{HE,DHW,*}, u_{HE,*max} \leftarrow 1$ 
4: else if  $R_{DHW} = 1$  then
5:    $\vartheta_{TES,DHW,min} \leftarrow \underline{\vartheta}_{TES,DHW}$ 
6: else if  $R_{SH} = 2$  then
7:    $u_{HP,SH,min} \leftarrow 1$ 
8:    $u_{HE,SH,*}, u_{HE,*max} \leftarrow 1$ 
9: else if  $R_{SH} = 1$  then
10:   $u_{HP,SH,min} \leftarrow 1$ 
11: end if

```

A.4. Enhanced heuristics

A.4.1. Electrical subsystem

The electrical subsystem enhancement implements a battery SoC reserve strategy based on PV generation as well as load and standby consumption forecasts, representing a compromise between control complexity and curtailment reduction while preserving self-consumption maximization. The required reserve calculation, as expressed in the following set of equations, determines the battery capacity needed to accommodate as much as possible predicted excess generation for the remainder of the day.

The first four steps follow the same principle as presented by Riesen, Ballif, and Wyrsh [101] and Litjens, Worrell, and Sark [102] for scheduling battery charging, adapted here with regard to the specific system architecture under consideration. The fifth step represents a logical advancement to also conservatively schedule discharging—as late as possible—in light of declining static feed-in limits or increasing number and length of periods without feed-in compensation.

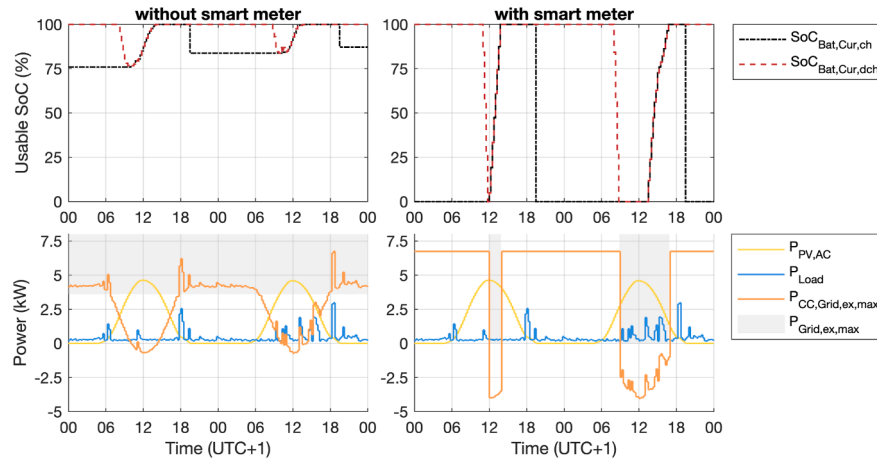


Fig. A.2. Upper charging and lower discharging limit using enhanced heuristics.

The procedure begins by calculating the inverter's AC power at maximum grid export $P_{Inv,AC,Grid,ex,max}$ through solving the AC node balance.

$$P_{Inv,AC,Grid,ex,max}(t) = -\max \left\{ \min \left\{ P_{PV,AC}(t) - P_{Load}(t) \right. \right. \\ \left. \left. - P_{Grid,ex,max}(t), \frac{P_{Inv,AC/DC,max}}{\eta_{Inv,AC/DC}} \right\}, \right. \\ \left. - \eta_{Inv,DC/AC} \cdot P_{Inv,DC/AC,max} \right\} \quad (A.34)$$

This value is then translated into its DC equivalent $P_{Inv,Grid,ex,max}$.

$$P_{Inv,Grid,ex,max}(t) = \eta_{Inv,AC/DC} \cdot \max \{ -P_{Inv,AC,Grid,ex,max}(t), 0 \} \\ - \frac{\min \{ -P_{Inv,AC,Grid,ex,max}(t), 0 \}}{\eta_{Inv,DC/AC}} \quad (A.35)$$

Subsequently, the charge controller's power at maximum grid export $P_{CC,Grid,ex,max}$ is derived by solving the DC node balance.

$$P_{CC,Grid,ex,max}(t) = -\max \left\{ \min \left\{ P_{PV,DC}(t) - P_{Stby}(t) \right. \right. \\ \left. \left. + P_{Inv,Grid,ex,max}(t), P_{CC,ch,max} \right\}, \right. \\ \left. - P_{CC,dch,max} \right\} \quad (A.36)$$

The upper charging limit $SoC_{Bat,Cur,ch}$ is determined by integrating the battery charging power at maximum grid export until the next sunset.

$$SoC_{Bat,Cur,ch}(t) = SoC_{Bat,max} \\ - \frac{\eta_{CC,ch} \cdot \int_{now}^{next\ sunset} \max \{ -P_{CC,Grid,ex,max}(t), 0 \} dt}{E_{Bat,max}} \quad (A.37)$$

Finally, the lower discharging limit $SoC_{Bat,Cur,dch}$ is computed by offsetting the charging limit with the integral of the battery's discharging power at maximum grid export until the next potential curtailment moment (when $P_{CC,Grid,ex,max} < 0$).

$$SoC_{Bat,Cur,dch}(t) = SoC_{Bat,Cur,ch}(t) \\ + \frac{\int_{now}^{next\ curtailment} \max \{ P_{CC,Grid,ex,max}(t), 0 \} dt}{\eta_{CC,dch} \cdot E_{Bat,max}} \quad (A.38)$$

Fig. A.2 exemplarily depicts the procedure's results for two consecutive days, illustrating the difference between operation with and without a smart meter.

Eq. (A.39) then lowers the effective SoC ceiling $SoC_{Bat,Cur}$, beyond which grid export becomes more attractive than storing energy.

$$SoC_{Bat,Cur}(t) = \min \{ \max \{ SoC_{Bat}(t), SoC_{Bat,Cur,ch}(t) \}, SoC_{Bat,Cur,dch}(t) \} \quad (A.39)$$

Since charging is still preferable to curtailment, the latter shall only occur when unavoidable, which is achieved through the introduction of decision variable $\Delta SoC_{Bat,Cur}$ with associated cost $c_{\Delta SoC,Bat,Cur}$, as defined in Inequalities (A.40) and (A.41) and by condition (A.42), respectively.

$$\Delta SoC_{Bat,Cur} \geq SoC_{Bat}(T) - SoC_{Bat,Cur}(T) \quad (A.40)$$

$$\Delta SoC_{Bat,Cur} \geq 0 \quad (A.41)$$

$$\frac{c_{P,CC,ch}}{\eta_{CC,ch}} < \frac{c_{\Delta SoC,Bat} - c_{\Delta SoC,Bat,Cur}}{E_{Bat,max}} < \frac{c_{P,CC,ch} - \frac{c_{P,Grid,ex}}{\eta_{Inv,AC/DC}}}{\eta_{CC,ch}} \quad (A.42)$$

A.4.2. Thermal subsystem

Surplus PV power is exploited thermally, further increasing self-consumption rates [21,26]. If a corresponding demand is predicted over the subsequent 24-hour period, heat pump operation is favored over grid feed-in within elevated hysteresis bands for both DHW and SH modes. The underlying operational logic adjustments are depicted in Algorithms 4 and 5.

Algorithm 4 Surplus driven space heat request ($R_{DHW,+}$ analogous).

Input: $\vartheta_{TES,SH,+}$, $\vartheta_{TES,SH}$, $u_{HP,*}$, $\int_{now}^{in\ 24\ h} \dot{Q}_{SH}(t) dt$
Output: $R_{SH,+}$
1: **static** $R_{SH,+} \leftarrow 0$
2: **if** $\vartheta_{TES,SH} \geq \vartheta_{TES,SH,max}$ **or** $\int_{now}^{in\ 24\ h} \dot{Q}_{SH}(t) dt \leq 0$ **then**
3: $R_{SH,+} \leftarrow 0$
4: **else if** $\vartheta_{TES,SH} \leq \vartheta_{TES,SH,+}$ **then**
5: $R_{SH,+} \leftarrow 1$
6: **else if** $u_{HP,*} = 0$ **then**
7: $R_{SH,+} \leftarrow 0$
8: **end if**

Algorithm 5 Enhanced one-step optimization problem adaption.**Input:** $c_{P,HP,+}$, $R_{DHW,+}$, $R_{SH,+}$, R_{SH} **Output:** $c_{P,HP,DHW}$, $c_{P,HP,SH}$

```

1: if  $R_{DHW,+} = 1$  and  $R_{SH} = 0$  then
2:    $c_{P,HP,DHW} \leftarrow c_{P,HP,+}$ 
3: else if  $R_{SH,+} = 1$  then
4:    $c_{P,HP,SH} \leftarrow c_{P,HP,+}$ 
5: end if

```

The above mentioned order of priority is ensured by assigning the surplus power related heat pump cost (actually revenues) according to condition (A.43).

$$-c_{P,Grid,ex} < -c_{P,HP,+} < \frac{\frac{\eta_{CC,ch} \cdot c_{\Delta SoC,Bat}}{E_{Bat,max}} - c_{P,CC,ch}}{\eta_{Inv,AC/DC}} \quad (A.43)$$

Moreover, the start-up costs need to be adjusted as stated in Eq. (A.44) below, where $\Delta P_{HP,+}$ represents a threshold value above minimum power which needs to be exceeded for the heat pump to start running in surplus mode, aiming to avoid short-cycling.

$$c_{v,HP} = \Delta t \cdot (c_{P,Grid,ex} - c_{P,HP,+}) \cdot \Delta P_{HP,+} \quad (A.44)$$

Appendix B. Results**B.1. Summary**

Tables B.1–B.3

Table B.1

Result summary (2025).

	w/o SM			w/ SM		
	RRBC	ERBC	EMPC	RRBC	ERBC	EMPC
Net cost in €	285	203 → -82	153 → -132	469	353 → -116	293 → -176
Grid import in kWh (€)	2804 (802)	2532 (724) → -272 (-78)	2436 (697) → -368 (-105)	2804 (802)	2533 (724) → -271 (-77)	2435 (697) → -369 (-105)
Battery throughput in kWh (€)	2455 (61)	2262 (57) → -193 (-5)	2081 (52) → -373 (-9)	2455 (61)	2610 (65) → +155 (+4)	2344 (59) → -110 (-3)
Heat pump cycles (€)	889 (27)	848 (25) → -41 (-1)	651 (20) → -238 (-7)	889 (27)	821 (25) → -68 (-2)	662 (20) → -227 (-7)
Grid export in kWh (€)	7789 (-605)	7770 (-603) → -19 (+1)	7926 (-615) → +137 (-11)	5421 (-421)	5941 (-461) → +520 (-40)	6204 (-482) → +783 (-61)

Table B.2

Result summary (2024).

	w/o SM			w/ SM		
	RRBC	ERBC	EMPC	RRBC	ERBC	EMPC
Net cost in €	319	258 → -61	210 → -108	447	362 → -84	307 → -139
Grid import in kWh (€)	2821 (807)	2625 (751) → -196 (-56)	2549 (729) → -271 (-78)	2821 (807)	2630 (752) → -191 (-54)	2552 (730) → -269 (-77)
Battery throughput in kWh (€)	2311 (58)	2136 (53) → -175 (-4)	1939 (48) → -372 (-9)	2311 (58)	2387 (60) → +76 (+2)	2085 (52) → -226 (-6)
Heat pump cycles (€)	976 (29)	952 (29) → -24 (-1)	705 (21) → -271 (-8)	976 (29)	931 (28) → -45 (-1)	716 (21) → -260 (-8)
Grid export in kWh (€)	7405 (-575)	7404 (-575) → -1 (+0)	7577 (-588) → +172 (-13)	5759 (-447)	6151 (-478) → +392 (-30)	6392 (-496) → +633 (-49)

Table B.3

Result summary (2023).

	w/o SM			w/ SM		
	RRBC	ERBC	EMPC	RRBC	ERBC	EMPC
Net cost in €	431	368 → -63	318 → -113	475	412 → -64	357 → -119
Grid import in kWh (€)	3101 (887)	2903 (830) → -198 (-57)	2810 (804) → -291 (-83)	3101 (887)	2904 (831) → -197 (-56)	2812 (804) → -289 (-83)
Battery throughput in kWh (€)	2243 (56)	2070 (52) → -173 (-4)	1862 (47) → -381 (-10)	2243 (56)	2188 (55) → -55 (-1)	1915 (48) → -328 (-8)
Heat pump cycles (€)	1003 (30)	950 (29) → -53 (-2)	738 (22) → -265 (-8)	1003 (30)	942 (28) → -61 (-2)	730 (22) → -273 (-8)
Grid export in kWh (€)	6982 (-542)	6987 (-542) → +5 (+0)	7140 (-554) → +158 (-12)	6409 (-498)	6465 (-502) → +56 (-4)	6659 (-517) → +251 (-19)

B.2. Energy utilization

Figs. B.1, B.2

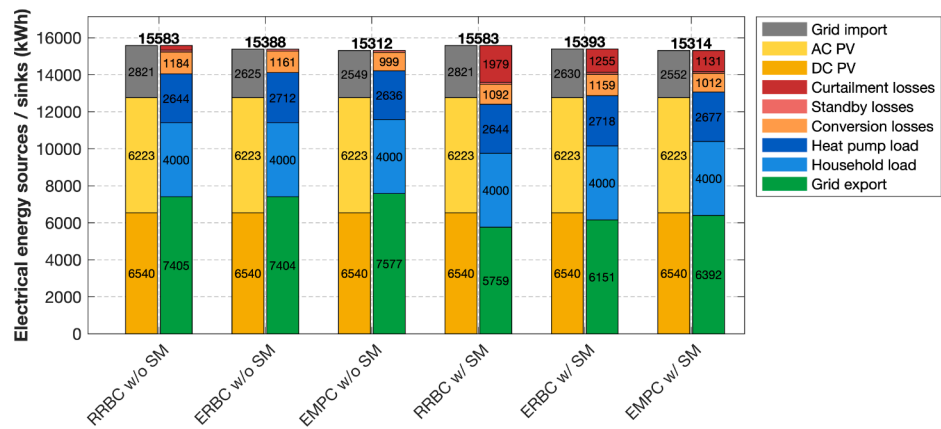


Fig. B.1. Annual electrical energy generation and consumption (2024).

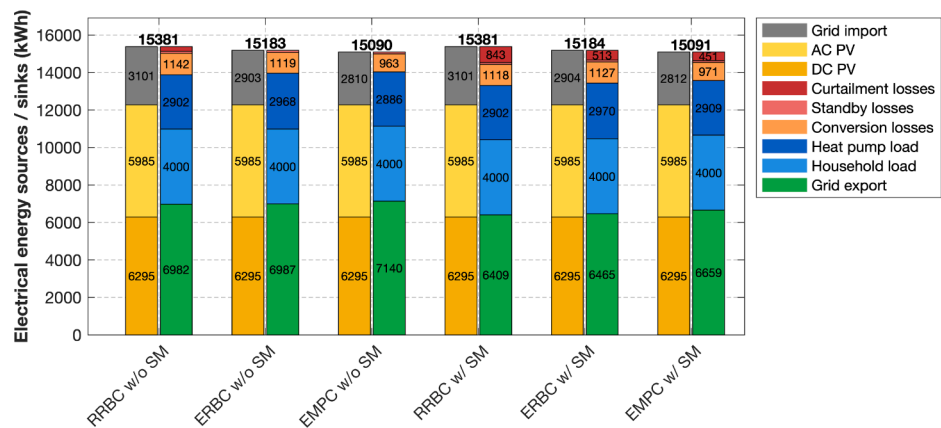


Fig. B.2. Annual electrical energy generation and consumption (2023).

B.3. Component degradation

Figs. B.3, B.4

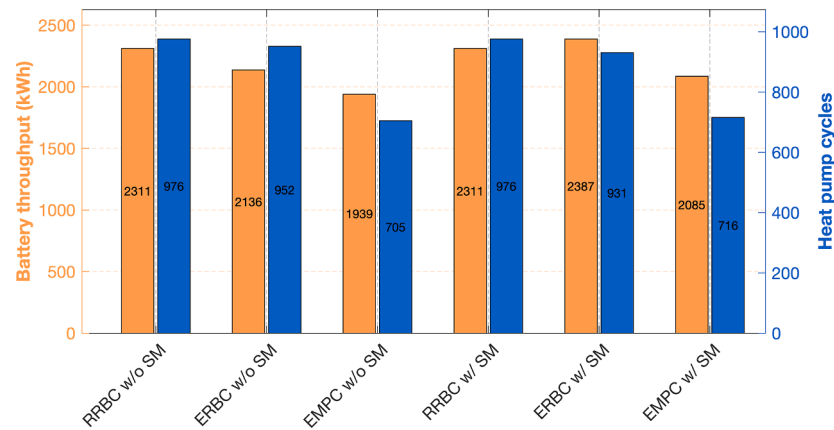


Fig. B.3. Annual component degradation parameters (2024).

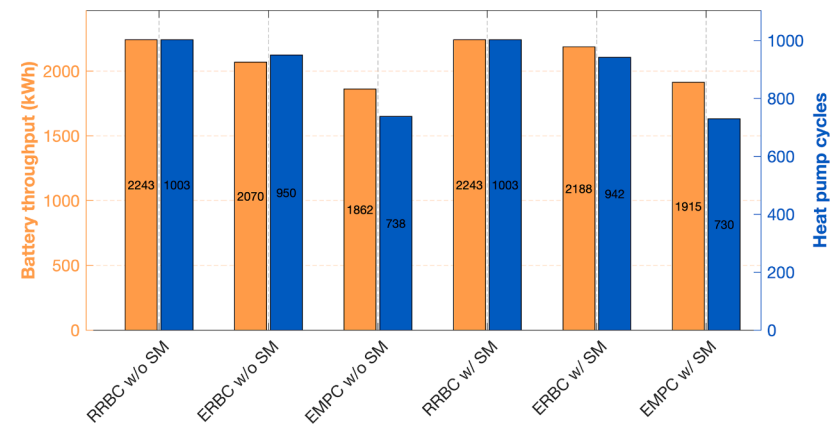


Fig. B.4. Annual component degradation parameters (2023).

B.4. Costs and revenues

Figs. B.5, B.6

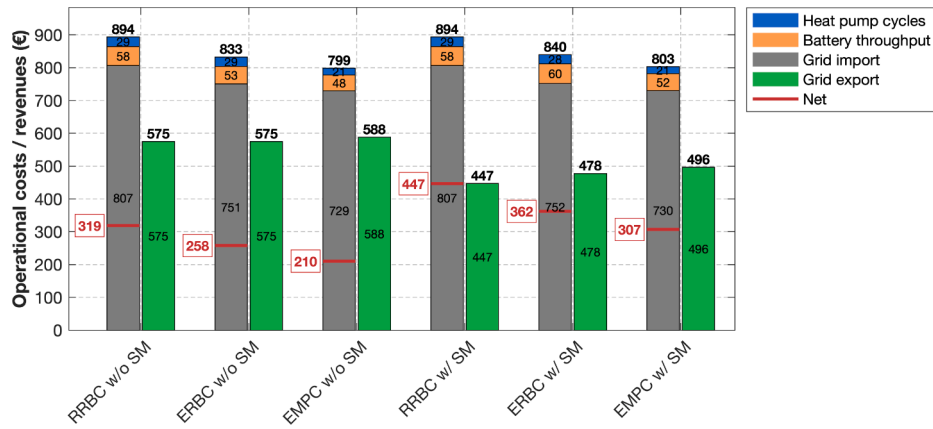


Fig. B.5. Costs and revenues (2024).

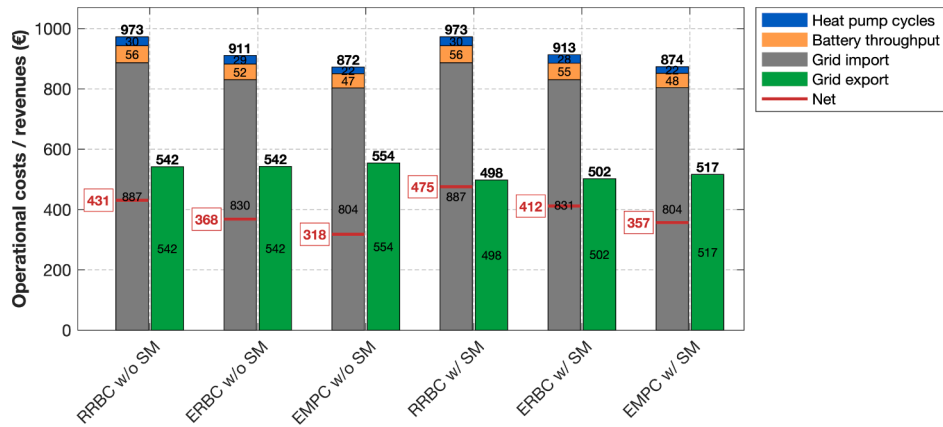


Fig. B.6. Costs and revenues (2023).

References

- [1] IPCC, Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, 2023, Intergovernmental Panel on Climate Change (IPCC), Geneva, Switzerland. Lee, H., Romero, J. (Eds.), Core Writing Team. <https://doi.org/10.59327/IPCC/AR6-9789291691647>
- [2] United Nations, Paris Agreement, 2015, adopted by the UNFCCC on December 12, 2015, https://unfccc.int/sites/default/files/resource/parisagreement_publication.pdf.
- [3] European Commission, A Renovation Wave for Europe, 2020. Accessed: 2023-10-27, https://energy.ec.europa.eu/system/files/2020-10/eu_renovation_wave_strategy_0.pdf.
- [4] B. Kroposki, Integrating high levels of variable renewable energy into electric power systems, J. Mod. Power Syst. Clean Energy 5 (6) (2017) 831–837. <https://doi.org/10.1007/s40565-017-0339-3>
- [5] Z. Lee, K. Gupta, K.J. Kircher, K.M. Zhang, Mixed-integer model predictive control of variable-speed heat pumps, Energy Build. 198 (2019) 75–83. <https://doi.org/10.1016/j.enbuild.2019.05.060>
- [6] E. Buechler, A. Goldin, R. Rajagopal, Designing model predictive control strategies for grid-interactive water heaters for load shifting applications, Appl. Energy 382 (2025) 125149. <https://doi.org/10.1016/j.apenergy.2024.125149>
- [7] L. Moretti, E. Martelli, G. Manzolini, An efficient robust optimization model for the unit commitment and dispatch of multi-energy systems and microgrids, Appl. Energy 261 (2020) 113859. <https://doi.org/10.1016/j.apenergy.2019.113859>
- [8] A. Parisio, C. Del Vecchio, A. Vaccaro, A robust optimization approach to energy hub management, Int. J. Electr. Power Energy Syst. 42 (1) (2012) 98–104. <https://doi.org/10.1016/j.ijepes.2012.03.015>
- [9] H. Wang, K. Meng, Z.Y. Dong, Z. Xu, F. Luo, K.P. Wong, Efficient real-time residential energy management through MILP based rolling horizon optimization, in: 2015 IEEE Power & Energy Society General Meeting, IEEE, 2015, pp. 1–6. <https://doi.org/10.1109/pesgm.2015.7285754>
- [10] J. Moshövel, K.-P. Kairies, D. Magnor, M. Leuthold, M. Bost, S. Gähres, E. Szczechowicz, M. Cramer, D.U. Sauer, Analysis of the maximal possible grid relief from PV-peak-power impacts by using storage systems for increased self-consumption, Appl. Energy 137 (2015) 567–575. <https://doi.org/10.1016/j.apenergy.2014.07.021>
- [11] S. Quoilin, K. Kavvadias, A. Mercier, I. Pappone, A. Zucker, Quantifying self-consumption linked to solar home battery systems: statistical analysis and economic assessment, Appl. Energy 182 (2016) 58–67. <https://doi.org/10.1016/j.apenergy.2016.08.077>
- [12] G.B.M.A. Litjens, E. Worrell, W.G. J. H.M. van Sark, Economic benefits of combining self-consumption enhancement with frequency restoration reserves provision by photovoltaic-battery systems, Appl. Energy 223 (2018) 172–187. <https://doi.org/10.1016/j.apenergy.2018.04.018>
- [13] H.J.J. Yu, System contributions of residential battery systems: new perspectives on PV self-consumption, Energy Econ. 96 (2021) 105151. <https://doi.org/10.1016/j.eneco.2021.105151>
- [14] Bundesministerium für Wirtschaft und Technologie, Erneuerbare-Energien-Gesetz, BGBl. 2012 I S. 1754, 2012. Accessed: 2026-08-07, https://www.clearingstelle-eeg-kwkg.de/sites/default/files/EEG2012_juris_120817.pdf.
- [15] Bundesministerium für Wirtschaft und Klimaschutz, Gesetz zur Änderung des Energiewirtschaftsrechts zur Vermeidung von temporären Erzeugungsüberschüssen, BGBl. 2025 I Nr. 51, 2025. Accessed: 2026-08-07, https://www.clearingstelle-eeg-kwkg.de/sites/default/files/EEG2025_I_Nr_51.pdf.
- [16] The European Parliament and The Council of the European Union, Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources (recast), 2018, (Official Journal of the European Union, L 328, 21 December 2018, pp. 82–209). EUR-Lex; accessed 2026-03-19, <https://eur-lex.europa.eu/eli/dir/2018/2001/oj/eng>.
- [17] The European Parliament and The Council of the European Union, Directive (EU) 2023/2413 of the European Parliament and of the Council of 18 October 2023

- amending Directive (EU) 2018/2001, Regulation (EU) 2018/1999 and Directive 98/70/EC as regards the promotion of energy from renewable sources, and repealing Council Directive (EU) 2015/652, 2023, (Official Journal of the European Union, L 2023/2413, 31 October 2023). EUR-Lex; accessed 2026-03-19, <https://eur-lex.europa.eu/eli/dir/2023/2413/oj/eng>.
- [18] A. Zeh, R. Witzmann, Operational strategies for battery storage systems in low-voltage distribution grids to limit the feed-in power of roof-mounted solar power systems, *Energy Procedia* 46 (2014) 114–123. <https://doi.org/10.1016/j.egypro.2014.01.164>
 - [19] T. Haupt, K. Settler, J. Jungwirth, H. Vaidya, Home-Energy-Management-Systeme (HEMS): Ein Marktüberblick für Deutschland, in: Tagungsband 39. PV-Symposium / BIPV-Forum, 2024.
 - [20] J. Weniger, N. Orth, L. Meissner, R. Schreier, N. Munzke, Stromspeicher-Inspektion 2025, Technical Report, 2025.
 - [21] D. Fischer, J. Bernhardt, H. Madani, C. Wittwer, Comparison of control approaches for variable speed air source heat pumps considering time variable electricity prices and PV, *Appl. Energy* 204 (2017) 93–105. <https://doi.org/10.1016/j.apenergy.2017.06.110>
 - [22] J. Clauß, L. Georges, Model complexity of heat pump systems to investigate the building energy flexibility and guidelines for model implementation, *Appl. Energy* 255 (2019) 113847. <https://doi.org/10.1016/j.apenergy.2019.113847>
 - [23] M. Evens, A. Arteconi, Influence of internal control simplifications in heat pump system modelling for energy flexibility evaluations, in: Proceedings of Building Simulation 2021: 17th Conference of IBPSA, BS 2021, KU Leuven, 2021. <https://doi.org/10.26868/25222708.2021.30214>
 - [24] S. Göbel, K. Waiza, C. Vering, D. Müller, The impact of controller settings in heat pumps: numerical findings and experimental verification, in: 36th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS 2023), Las Palmas de Gran Canaria, Spain, 2023. <https://doi.org/10.52202/069564-0075>
 - [25] T. Tjaden, J. Weniger, J. Bergner, V. Quaschnig, Wärmewende - Wärmepumpen und der Einfluss von PV-Batteriespeichern auf einen netzdienlichen Betrieb, in: 1. Herbstworkshop der Professur Energiespeichersysteme, Dezentrale Sektorkopplung und Hybride Energiespeichersysteme, HTW Berlin, 2016.
 - [26] J. Wiedera, J. Holdorf, Solarstromoptimiertes Energiemanagement von Wärmepumpen, Technical Report, 2023.
 - [27] I. Abdou, M. Tkouat, Unit commitment problem in electrical power system: a literature review, *Int. J. Electr. Comput. Eng.* 8 (3) (2018) 1357. <https://doi.org/10.11591/ijece.v8i3.pp1357-1372>
 - [28] L. Montero, A. Bello, J. Reneses, A review on the unit commitment problem: approaches, techniques, and resolution methods, *Energies* 15 (4) (2022) 1296. <https://doi.org/10.3390/en15041296>
 - [29] Á.O. Topa Gavilema, J.D. Álvarez, J.L. Torres Moreno, M.P. García, Towards optimal management in microgrids: an overview, *Energies* 14 (16) (2021) 5202. <https://doi.org/10.3390/en14165202>
 - [30] G.S. Thirunavukkarasu, M. Seyedmahmoudian, E. Jamei, B. Horan, S. Mekhilef, A. Stojcevski, Role of optimization techniques in microgrid energy management systems-a review, *Energy Strategy Rev.* 43 (2022) 100899. <https://doi.org/10.1016/j.esr.2022.100899>
 - [31] S. Ahmad, M. Shafiqullah, C.B. Ahmed, M. Alowafeer, A review of microgrid energy management and control strategies, *IEEE Access* 11 (2023) 21729–21757. <https://doi.org/10.1109/access.2023.3248511>
 - [32] Z. Wang, T. Hong, M.A. Piette, Building thermal load prediction through shallow machine learning and deep learning, *Appl. Energy* 263 (2020) 114683. <https://doi.org/10.1016/j.apenergy.2020.114683>
 - [33] P. Ma, S. Cui, M. Chen, S. Zhou, K. Wang, Review of family-level short-term load forecasting and its application in household energy management system, *Energies* 16 (15) (2023) 5809. <https://doi.org/10.3390/en16155809>
 - [34] C. Scott, M. Ahsan, A. Albarbar, Machine learning for forecasting a photovoltaic (PV) generation system, *Energy* 278 (2023) 127807. <https://doi.org/10.1016/j.energy.2023.127807>
 - [35] C. Baumann, P. Wohlgenannt, W. Streicher, P. Kepplinger, Optimizing heat pump control in an NZEB via model predictive control and building simulation, *Energies* 18 (1) (2024) 100. <https://doi.org/10.3390/en18010100>
 - [36] C. Betzold, S. Bordin, A. Dentel, G. Harhausen, Control strategies for modulating heat pumps in a plus energy building, in: Proceedings of the 13th IEA Heat Pump Conference, Jeju, Korea, 2021. Paper No. 042, <https://heat-pumpingtechnologies.org/publications/paper-no-042-control-strategies-for-modulating-heat-pumps-in-a-plus-energybuilding-13th-iea-heat-pump-conference-jeju-korea/>.
 - [37] G. Bianchini, M. Casini, D. Pepe, A. Vicino, G.G. Zanvetto, An integrated model predictive control approach for optimal HVAC and energy storage operation in large-scale buildings, *Appl. Energy* 240 (2019) 327–340. <https://doi.org/10.1016/j.apenergy.2019.01.187>
 - [38] C. Corinaldesi, D. Schwabeneder, G. Lettner, H. Auer, A rolling horizon approach for real-time trading and portfolio optimization of end-user flexibilities, *Sustain. Energy Grids Netw.* 24 (2020) 100392. <https://doi.org/10.1016/j.segan.2020.100392>
 - [39] P. Fitzpatrick, F. D'Ettorre, M. De Rosa, M. Yadack, U. Eicker, D.P. Finn, Influence of electricity prices on energy flexibility of integrated hybrid heat pump and thermal storage systems in a residential building, *Energy Build.* 223 (2020) 110142. <https://doi.org/10.1016/j.enbuild.2020.110142>
 - [40] R. Gelleschus, T. Bocklisch, Impact of Model and Forecast Uncertainties on the Performance of the Model Predictive Control of a PV-Battery-Heat Pump-Heat Storage System, *Atlantis Press International BV*, 2023, pp. 162–192. https://doi.org/10.2991/978-94-6463-156-2_13
 - [41] H. Harb, Vorausschauende Demand Side Management Strategien für Wohngebäudeenergiesysteme, Ph.D. thesis, RWTH Aachen University, Aachen, 2017. <https://doi.org/10.18154/RWTH-2017-09952>
 - [42] S. Kuboth, F. Heberle, A. König-Haagen, D. Brüggemann, Economic model predictive control of combined thermal and electric residential building energy systems, *Appl. Energy* 240 (2019) 372–385. <https://doi.org/10.1016/j.apenergy.2019.01.097>
 - [43] L. Langer, T. Volling, An optimal home energy management system for modulating heat pumps and photovoltaic systems, *Appl. Energy* 278 (2020) 115661. <https://doi.org/10.1016/j.apenergy.2020.115661>
 - [44] R. Liu, Z. Liu, Y. Leng, B. Li, Y. Wu, Economic model predictive control for an office building with photovoltaic, heat pump, thermal storage and electric vehicle under time-of-use tariff, *Build. Environ.* 282 (2025) 113312. <https://doi.org/10.1016/j.buildenv.2025.113312>
 - [45] Y. Löhr, M. Mönnigmann, Optimal operation of electrical heating system with hybrid model predictive control, *IFAC-Pap.* 51 (28) (2018) 274–279. <https://doi.org/10.1016/j.ifacol.2018.11.714>
 - [46] L. Maier, M. Schönegge, S. Henn, D. Hering, D. Müller, Assessing mixed-integer-based heat pump modeling approaches for model predictive control applications in buildings, *Appl. Energy* 326 (2022) 119894. <https://doi.org/10.1016/j.apenergy.2022.119894>
 - [47] A. Murray, T. Schütz, G. Pan, D. Müller, V. Hagenmeyer, On modelling effects in the battery and thermal storage scheduling problem, *J. Build. Perform. Simul.* 14 (1) (2020) 38–51. <https://doi.org/10.1080/19401493.2020.1838611>
 - [48] S. Rafii-Tabrizi, J.-R. Hadji-Minaglou, F. Scholzen, F. Capitanescu, Optimal operation of nearly zero energy buildings using mixed integer linear programming, in: 2019 International Conference on Smart Energy Systems and Technologies (SEST), IEEE, 2019, pp. 1–6. <https://doi.org/10.1109/sest.2019.8849083>
 - [49] P. Sawant, O. Villegas Mier, M. Schmidt, J. Pfafferott, Demonstration of optimal scheduling for a building heat pump system using economic-MPC, *Energies* 14 (23) (2021) 7953. <https://doi.org/10.3390/en14237953>
 - [50] T. Schütz, H. Harb, R. Streblow, D. Müller, Comparison of models for thermal energy storage units and heat pumps in mixed integer linear programming, in: The 28th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems, ECOS 2015, 29.06.2015–03.07.2015, Pau, France, 2015. <https://publications.rwth-aachen.de/record/540090>.
 - [51] W. Tang, Y. Li, S. Walker, T. Keviczky, Model predictive control design for unlocking the energy flexibility of heat pump and thermal energy storage systems, 2024. <https://doi.org/10.48550/ARXIV.2402.12488>
 - [52] L. Tomás, M. Lämmle, J. Pfafferott, Demonstration and evaluation of model predictive control (MPC) for a real-world heat pump system in a commercial low-energy building for cost reduction and enhanced grid support, *Energies* 18 (6) (2025) 1434. <https://doi.org/10.3390/en18061434>
 - [53] C. Verhelst, F. Logist, J. Van Impe, L. Helsen, Study of the optimal control problem formulation for modulating air-to-water heat pumps connected to a residential floor heating system, *Energy Build.* 45 (2012) 43–53. <https://doi.org/10.1016/j.enbuild.2011.10.015>
 - [54] WAGO Kontakttechnik GmbH & Co. KG, Automation Gateway Modbus, Version 1.0.0, 2020.
 - [55] M. Vader, Data communication with Victron Energy products, Victron Energy B.V., 31st edition, 2021.
 - [56] SMA Solar Technology AG, SMA Modbus® Interface, Version 1.6, 2022. [EDMx-Modbus-TI-en-16](https://www.sma.de/fileadmin/user_upload/Dateien/Modbus-TI-en-16).
 - [57] G.A. Aragón Cabrera, Extended model predictive control software framework for real-time local management of complex energy systems, Ph.D. thesis, 2021. <https://doi.org/10.18154/RWTH-2021-02212>
 - [58] G. Schweiger, L.V. Eckerstorfer, I. Hafner, A. Fleischhacker, J. Radl, B. Glock, M. Wastian, M. Rößler, G. Lettner, N. Popper, K. Corcoran, Active consumer participation in smart energy systems, *Energy Build.* 227 (2020) 110359. <https://doi.org/10.1016/j.enbuild.2020.110359>
 - [59] T. Loga, B. Stein, N. Diefenbach, TABULA building typologies in 20 European countries-making energy-related features of residential building stocks comparable, *Energy Build.* 132 (2016) 4–12. <https://doi.org/10.1016/j.enbuild.2016.06.094>
 - [60] Deutsches Institut für Normung, DIN/TS 18599-10:2025-10: Energetische Bewertung von Gebäuden – Berechnung des Nutz-, End- und Primärenergiebedarfs für Heizung, Kühlung, Lüftung, Trinkwarmwasser und Beleuchtung – Teil 10: Nutzungsrandbedingungen, Klimadaten, Technische Spezifikation, Beuth Verlag, Berlin, 2025.
 - [61] Verein Deutscher Ingenieure, VDI 4645:2023-07: Heizungsanlagen mit Wärmepumpen in Ein- und Mehrfamilienhäusern – Planung, Errichtung, Betrieb, Richtlinie, Beuth Verlag, Berlin, 2023.
 - [62] Deutsches Institut für Normung, DIN EN 12831-1:2017-09: Energetische Bewertung von Gebäuden – Verfahren zur Berechnung der Norm-Heizlast – Teil 1: Raumheizlast, Modul M3-3, Beuth Verlag, Berlin, 2017. Deutsche Fassung EN 12831-1:2017.
 - [63] Trägerschaft Wärmepumpen-System-Modul, Dimensionierungsvorgaben Speicher im WPSM-Pflichtenheft mit und ohne Eigenstromnutzung, 2018. Accessed: 2026-08-07, <https://www.wp-systemmodul.ch/files/Dimensionierungsvorgaben-Speicher-180810.pdf>.
 - [64] Heizungsanlagen in Gebäuden – Planung von Heizungsanlagen mit Wärmepumpen, 2007, Deutsche Fassung EN 15450:2007.
 - [65] N. Orth, J. Weniger, L. Meissner, Empfehlungen zur Auslegung von Solarstromspeichern, *Sonnenenergie* (2) (2022) 16–17. Forschungsgruppe Solarspeichersysteme.
 - [66] Deutscher Wetterdienst, DWD Open Data Server, 2025. Accessed: 2025-10-31, <https://opendata.dwd.de/>.

- [67] W.F. Holmgren, C.W. Hansen, M.A. Mikowski, pvlib python: a python package for modeling solar energy systems, *J. Open Source Softw.* 3 (29) (2018) 884. <https://doi.org/10.21105/joss.00884>
- [68] International Organization for Standardization, ISO 13790:2008: Energy performance of buildings – Calculation of energy use for space heating and cooling, Standard, Geneva, Switzerland, 2008.
- [69] M. Lange, M. Zobel, Schlussbericht zum Vorhaben: Erstellung neuer Referenzlastprofile zur Auslegung, Dimensionierung und Wirtschaftlichkeitsberechnung von Hausenergieversorgungssystemen mit dem Akronym NOVAREF : im Rahmen der Fördermaßnahme: Transfer von Forschungs- und Entwicklungsergebnissen *FuE* durch Normung und Standardisierung, Technical Report, Oldenburg, 2017. <https://www.tib.eu/de/suchen/id/TIBKAT%3A890929130>.
- [70] Verein Deutscher Ingenieure, VDI 4655:2021-11: Referenzlastprofile von Wohngebäuden für Strom, Heizung und Trinkwarmwasser sowie Referenzerzeugungsprofile für Fotovoltaikanlagen, Richtlinie, Beuth Verlag, Berlin, 2021. <https://www.dinmedia.de/de/technische-regel/vdi-4655/336886496>.
- [71] Verivox GmbH, Strompreisentwicklung 2025 in Deutschland, 2025. Datenquelle: Verivox-Haushaltspreis Strom, <https://www.verivox.de/strom/strompreisentwicklung/>.
- [72] Bundesnetzagentur, SMARD – Strommarktdaten, 2025. Accessed: 2025-08-01, <https://www.smard.de/>.
- [73] R. Fahrlich, S. Brandt, M. Kriegel, N. Kießling, A. Studnior, F. Petrakopoulou, M. Albert, D. Wolf, Schlussbericht des Verbundvorhabens EG2050: FlexEhome - Das netzdienliche Solarhaus mit thermischer und elektrischer Vollversorgung, Projektabschlussbericht, 2026. Im Rahmen des 7. Energieforschungsprogramms der Bundesregierung, Förderschwerpunkt Energieeffiziente Gebäude – Innovationsprojekte, Förderkennzeichen 03EN3062A/B/C.
- [74] T. Beck, H. Kondziella, G. Huard, T. Bruckner, Assessing the influence of the temporal resolution of electrical load and PV generation profiles on self-consumption and sizing of PV-battery systems, *Appl. Energy* 173 (2016) 331–342. <https://doi.org/10.1016/j.apenergy.2016.04.050>
- [75] A. Burgio, D. Menniti, N. Sorrentino, A. Pinnarelli, Z. Leonowicz, Influence and impact of data averaging and temporal resolution on the assessment of energetic, economic and technical issues of hybrid photovoltaic-battery systems, *Energies* 13 (2) (2020) 354. <https://doi.org/10.3390/en13020354>
- [76] A. Vasylyev, A. Vannoni, A. Sorce, Optimal dispatch of Li-Ion battery energy storage, reviewing and considering cycling and calendar ageing models, *Appl. Therm. Eng.* 265 (2025) 125597. <https://doi.org/10.1016/j.applthermaleng.2025.125597>
- [77] Y. Löhr, M. Mönnigmann, Receding horizon heat flow control in domestic buildings – Yesterday-based disturbance predictions, in: 2018 European Control Conference (ECC), IEEE, 2018, pp. 404–409. <https://doi.org/10.23919/ecc.2018.8550375>
- [78] Y. Yoo, S. Jung, Modeling forecast errors for microgrid operation using Gaussian process regression, *Sci. Rep.* 14 (1) (2024). <https://doi.org/10.1038/s41598-024-52224-y>
- [79] F. Moazzen, M.J. Hossain, L. Li, B. Mohammadi-ivatloo, Energy management under uncertainty for hybrid microgrids: from data to decision-making, *IET Renew. Power Gener.* 19 (1) (2025). <https://doi.org/10.1049/rpg2.70174>
- [80] P.L.C. García-Miguel, J. Alonso-Martínez, S. Arnaltes Gómez, M. García Plaza, A.P. Asensio, A review on the degradation implementation for the operation of battery energy storage systems, *Batteries* 8 (9) (2022) 110. <https://doi.org/10.3390/batteries8090110>
- [81] L. Steijger, R. Buswell, V. Smedley, S. Firth, P. Firth, An air source heat pump model for operation in cold humid environments, in: 8th International Conference on System Simulation in Buildings, Liège, Belgium, 13th–15th December, 2010. https://repository.lboro.ac.uk/articles/conference_contribution/An_air_source_heat_pump_model_for_operation_in_cold_humid_environments/9434558.
- [82] EEBUS Initiative e.V., EEBUS Use Cases Overview (V1.7), 2022. Version 1.7, 22 December 2022, <https://www.eebus.org/wp-content/uploads/2023/04/20221222-EEBUS-Overview-Use-Cases-v1.7.pdf>.
- [83] M. Kim, T. Park, J. Jeong, H. Kim, Stochastic optimization of home energy management system using clustered quantile scenario reduction, *Appl. Energy* 349 (2023) 121555. <https://doi.org/10.1016/j.apenergy.2023.121555>
- [84] J.M. Maciejowski, *Predictive Control with Constraints*, Prentice Hall, Reprint, Harlow, 2008.
- [85] G. Morales-España, J.M. Latorre, A. Ramos, Tight and compact MILP formulation of start-up and shut-down ramping in unit commitment, *IEEE Trans. Power Syst.* 28 (2) (2013) 1288–1296. <https://doi.org/10.1109/tpwrs.2012.2222938>
- [86] B. Knueven, J. Ostrowski, J.-P. Watson, On mixed-integer programming formulations for the unit commitment problem, *INFORMS J. Comput.* (2020). <https://doi.org/10.1287/ijoc.2019.0944>
- [87] H. Madani, J. Claesson, P. Lundqvist, A descriptive and comparative analysis of three common control techniques for an on/off controlled ground source heat pump (GSHP) system, *Energy Build.* 65 (2013) 1–9. <https://doi.org/10.1016/j.enbuild.2013.05.006>
- [88] M.A. Bianchi, Adaptive modellbasierte prädiktive Regelung einer Kleinwärmepumpenanlage, Ph.D. thesis, ETH Zürich, 2006. <https://doi.org/10.3929/ETHZ-A-005302150>
- [89] D.L. Gerber, V. Vossos, W. Feng, C. Marnay, B. Nordman, R. Brown, A simulation-based efficiency comparison of AC and DC power distribution networks in commercial buildings, *Appl. Energy* 210 (2018) 1167–1187. <https://doi.org/10.1016/j.apenergy.2017.05.179>
- [90] B. Schaden, T. Jatschka, S. Limmer, G.R. Raidl, Smart charging of electric vehicles considering SOC-dependent maximum charging powers, *Energies* 14 (22) (2021) 7755. <https://doi.org/10.3390/en14227755>
- [91] European Committee for Standardization (CEN), EN 14511-2:2022: Air conditioners, liquid chilling packages and heat pumps for space heating and cooling and process chillers, with electrically driven compressors - Part 2: Test conditions, 2022.
- [92] A. Dentel, C. Kendel, T. Afjei, M. Wetter, TRNSYS TYPE 402, Transsolar Energietechnik GmbH, 2022.
- [93] H.P. Williams, *Model Building in Mathematical Programming*, 5th edition, John Wiley & Sons, Chichester, UK, 2013.
- [94] Tecalor GmbH, TTL ACS Luft/Wasser-Wärmepumpe: Bedienung und Installation, n.d.. Accessed: 2026-08-07, https://www.tecalor.de/toolbox/content/docs/anleitungen/installation/TTL_ACS/325926-41128-9437_TTL_3.5-8.5_ACS_de.pdf.
- [95] Wolf GmbH, Wolf-Regelungssysteme WRS WRS-2: Kurzanleitung, 2017. Accessed: 2026-08-07, <https://www.wolf.eu/de-de/professional/downloads/downloadcenter>.
- [96] CTC/Enertech AB, EcoLogic M/L: Installations- und Wartungsanleitung, 2023. Accessed: 2026-08-07, https://ctc-expert.ch/cm4all/uproc.php/0/Regler/DE_User%20Manual-CTC%20EcoLogic%20L_M-16250661.pdf?cdp=a&=1906ef88c48.
- [97] D. Spiers, Batteries in PV Systems, Elsevier, 2018, pp. 789–843. <https://doi.org/10.1016/b978-0-12-809921-6.00021-5>
- [98] R. Borah, F.R. Hughson, J. Johnston, T. Nann, On battery materials and methods, *Mater. Today Adv.* 6 (2020) 100046. <https://doi.org/10.1016/j.mtaadv.2019.100046>
- [99] Bundesverband Wärmepumpe e.V., Wie hoch ist die Lebensdauer einer Wärmepumpe? Wie kann ich diese erhöhen?, 2018, Accessed: 2025-05-20, <https://www.waermepumpe.de/waermepumpe/darum-waermepumpe/fragen-sie-die-experten/moegliche-fragen/wie-hoch-ist-die-lebensdauer-einer-waermepumpe-wie-kann-ich-diese-erhoehen/>.
- [100] Copeland LP, Compressor Short Cycling – An Unrecognized Problem, Application Engineering Bulletin AE17-1262 R2, 2024. Accessed: 2026-08-07, https://webapps.copeland.com/online-product-information/Publication/LaunchPDF?Index=aeb&PDF=AE17-1262_R2.pdf.
- [101] Y. Riesen, C. Ballif, N. Wyrsh, Control algorithm for a residential photovoltaic system with storage, *Appl. Energy* 202 (2017) 78–87. <https://doi.org/10.1016/j.apenergy.2017.05.016>
- [102] G.B.M.A. Litjens, E. Worrell, W.G. J. H.M. van Sark, Assessment of forecasting methods on performance of photovoltaic-battery systems, *Appl. Energy* 221 (2018) 358–373. <https://doi.org/10.1016/j.apenergy.2018.03.154>